MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: Mansourah Massarah Gold Project
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION
3210 GRINDING & CLASSIFICATION
DESIGN OF RETAINING WALL

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TABLE OF CONTENTS

DEFINITIONS AND ABBREVIATIONS 3
1. INTENT 4
2. EXECUTIVE SUMMARY 4
3. SCOPE 4
4. CONCRETE PARAMETERS 5
5. REINFORCEMENT PARAMETERS 5
6. LOAD COMBINATIONS 5
7. DESIGN OF RETAINING WALL TYPE-I 6
8. DESIGN OF RETAINING WALL TYPE-II 14

References 22
ATTACHMENT 1 MECHANICAL INPUT DRAWINGS 23
ATTACHMENT 2 CIVIL DRAWINGS 26

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
STAAD	:	Commonly used analysis and design software package named STAAD.Pro
Sec.	:	Section
max.	:	Maximum
ref.	:	Reference
No.	:	Number
cl.	:	Clause
nos.	:	Numbers
Col.	:	Column
approx.	:	Approximate
eq.	:	Equation
mkd.	:	Marked
Dia.	:	Diameter
Spac.	:	Spacing
Rebar	:	Reinforcement Bar

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1. INTENT

This document is to Prepare for design of Retaining Wall near **CRUSHED ORE STORAGE BIN (3210)**. The analysis & design is carried out in accordance with project specifications along with codes & standards.
The design calculation ensures the safety and serviceability of the complete retaining wall system.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel.
The stresses due to all possible load case combinations are calculated and used directly as force values for member strength calculation.

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4. CONCRETE PARAMETERS:

Type of concrete used	=	C35		
f_c'	=	35	Mpa	[i.e., 28 days compressive strength]
Unit weight of RCC	=	24	KN/m^3	[as per document number: MD-216-1000-EG-ST-CRT-1001]

5. REINFORCEMENT PARAMETERS:

Type of Steel used	=	ASTM A 615/ A 615M Grade 60
f_y	=	420 Mpa

6. LOAD COMBINATIONS:

STABILITY

1 D+H+W
2 D+L+H+W
3 D+H+0.7E
4 D+L+H+0.7E

STRENGTH

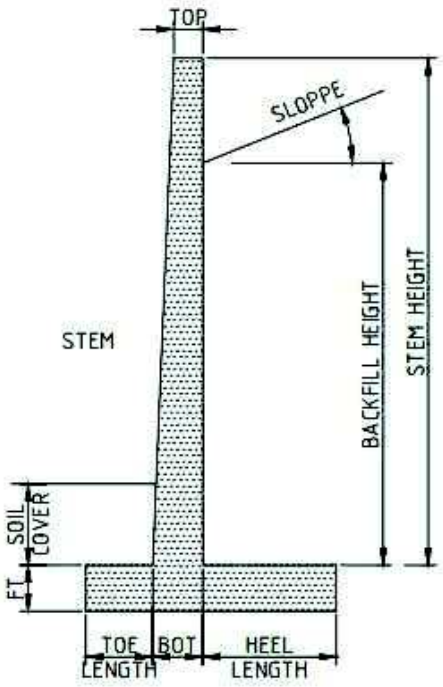
1 1.4D
2 1.2D+1.6(L+H)
3 1.2D+0.8W
4 1.2D+L+1.6W
5 1.2D+L+E
6 0.9D+1.6H+1.6W
7 0.9D+1.6H+E

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7. DESIGN OF RETAINING WALL TYPE-I:



Conc. Stem Height	7.50	m	
Stem Thickness Top	43.2	cm	
Stem Thickness Bot	68.7	cm	
Footing Thickness	70.0	cm	
Toe Length	1.06	m	
Heel Length	2.00	m	
Soil Cover @ Toe	0.30	m	
Backfill Height	6.70	m	OK
Backfill Slope Angle	10.0	deg	
Wind Load on Stem	0.9	KPa	
Wind Height from Top	7.20	m	

Total Base width = (2.0+0.687+1.063) = 3.75m; Height = (6.7+0.7)= 7.4m;
Ratio = (3.75 / 7.4) = 0.51; OK [as, 0.5< 0.51 < 0.7]

BACKFILL PROPERTIES

Wall taper α = 0.000 rad

Backfill slope $\beta = slope * \pi / 180$ = 10.0 * 3.14 / 180 = 0.175 rad

Internal friction $\phi = Int.\ friction * \pi / 180$ = 33.0 * 3.14 / 180 = 0.576 rad

Wall-soil friction $\delta = \phi / 2$ = 0.576 / 2 = 0.288 rad

Seismic angle $\theta = aTan\ (kh / (1 - kv))$ = aTan (0.02 / (1 - 0.02)) = 0.020 rad

Footing length $ftg = toe + stem + heel$ = 1.06 + 68.7 / 100 + 2.00 = 3.75 m

Height for Stability $Hs = wedge + backfill + footing$ = 0.35 + 6.70 + 70.0 / 100 = 7.75 m

Earth pressure theory = Coulomb Active Moist density = 19 KN/m³ Saturated density = 19 KN/m³

Active coefficient $ka = \dfrac{Cos^2(\phi + \alpha)}{Cos^2\alpha * Cos(\delta + \alpha) * [1 + \sqrt{\dfrac{Sin(\phi + \delta) * Sin(\phi - \beta)}{Cos(\delta + \alpha) * Cos(\beta - \alpha)}}\ ]^2}$ = 0.30

Active pressure $pa = ka * \gamma$ = 0.30 * 19.0 = 5.7 KPa/m of height

- For stability analysis (non-seismic)

Active force $Pa = ka * \gamma * Hs^2 / 2$ = 0.30 * 19.0 * 7.75² / 2 = 171.9 KN/m

$Pah = Pa * Cos(\delta + \alpha)$ = 171.9 * Cos(0.288 + 0.000) = 164.8 KN/m

$Pav = Pa * Sin(\delta + \alpha)$ = 171.9 * Sin(0.288 + 0.000) = 48.8 KN/m

Water force $Pw = (ka * (\gamma s - \gamma w - \gamma) + \gamma w) * (Water\ table)^2 / 2$

Pw = (0.30 * (19.0 - 9.8 - 19.0) + 9.8) * 0.73² / 2 = 1.8 KN/m

- For stem design (non-seismic)

Active force $Pa = ka * \gamma * H^2 / 2$ = 0.30 * 19.0 * 6.70² / 2 = 128.4 KN/m

$Pah = Pa * Cos(\delta + \alpha)$ = 128.4 * Cos (0.288 + 0.000) = 123.1 KN/m

$Pav = Pa * Sin(\delta + \alpha)$ = 128.4 * Sin (0.288 + 0.000) = 36.5 KN/m

Water force $Pw = (ka * (\gamma s - \gamma w - \gamma) + \gamma w) * (Water\ table - Ftg)^2 / 2$

Pw = (0.30 * (19.0 - 9.8 - 19.0) + 9.8) * (0.73 - 70.0 / 100)² / 2 = 0.0 KN/m

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Hor. seismic coeff. kh = 0.02 Ver. seismic coeff. kv = 0.02

Active seismic coeff. $kae = \frac{Cos^2(\phi - \alpha - \theta)}{Cos\theta * Cos^2\alpha * Cos(\delta + \alpha + \theta) * [1 + (\frac{Sin(\phi + \delta) * Sin(\phi - \beta - \theta)}{Cos(\delta + \alpha + \theta) * Cos(\beta - \alpha)})^{1/2}]^2} = 0.32$

- For stability analysis (seismic)

Seismic force $Pae = kae * \gamma * Hs^2 / 2 * (1 - kv) =$ 0.32 * 19.0 * 7.75² / 2 * (1 - 0.0) = 176.7 KN/m

$Paeh = Pae * Cos(\delta + \alpha) =$ 176.7 * Cos (0.288 + 0.000) = 169.4 KN/m

$Paev = Pae * Sin(\delta + \alpha) =$ 176.7 * Sin (0.288 + 0.000) = 50.2 KN/m

Water force $Pwe = kh * (\gamma s - \gamma) * (water)^2 / 2$

Pwe = 0.02 * (19.0 - 19.0) * (0.73)² / 2 = 0.0 KN/m

- For stem design (seismic)

Seismic force $Pae = kae * \gamma * H^2 / 2 =$ 0.32 * 19.0 * 6.70² / 2 = 132.0 KN/m

$Paeh = Pae * Cos(\delta + \alpha) =$ 132.0 * Cos (0.288 + 0.000) = 126.6 KN/m

$Paev = Pae * Sin(\delta + \alpha) =$ 132.0 * Sin (0.288 + 0.000) = 37.5 KN/m

Water force $Pwe = kh * (\gamma s - \gamma) * (water\ table - ftg)^2 / 2$

Pwe = 0.02 * (19.0 - 19.0) * (0.73 - 70.0 / 100)² / 2 = 0.0 KN/m

OVERTURNING CALCULATIONS (Comb. D+H+W)

- Overturning Forces

Backfill = $Lat\ factor * Pah =$ 1.0 * 164.8 = 164.8 KN/m

Arm = $Hs / 3 =$ 7.75 / 3 = 2.58 m Moment = 164.8 * 2.58 = 426.0 KN-m/m

Water table = $Lat\ factor * Pw =$ 1.0 * 1.8 = 1.8 KN/m

Arm = $Water\ table / 3 =$ 0.73 / 3 = 0.24 m Moment = 1.8 * 0.24 = 0.4 KN-m/m

Surcharge = $Lat\ factor * ka * Surcharge * Hs =$ 1.0 * 0.31 * 0.0 * 7.75 = 0.0 KN/m

Arm = $Hs / 2 =$ 7.75 / 2 = 3.88 m Moment = 0.0 * 3.88 = 0.0 KN-m/m

Strip load = $\Sigma\ Lat\ factor * 2 * Q / \pi * [\beta - Sin\beta * Cos(2\alpha)] =$ 0.0 KN/m

Arm = 3.35 m Moment = 0.0 * 3.35 = 0.0 KN-m/m

Wind load = $WL\ factor * Pressure * Wind\ height =$ 1.0 * 0.9 * 7.20 = 6.5 KN/m

Arm = $Ftg + Stem - Wind\ height / 2 =$ 70.0 / 100 + 7.50 - 7.20 / 2 = 4.60 m

Moment = 6.5 * 4.60 = 29.8 KN-m/m

Backfill seismic = $EQ\ factor * (Paeh - Pah) =$ 0.0 * (169.4 - 164.8) = 0.0 KN/m

Arm = $0.6 * Hs =$ 0.6 * 7.75 = 4.65 m Moment = 0.0 * 4.65 = 0.0 KN-m/m

Water seismic = $EQ\ factor * Pwe =$ 0.0 * 0.0 = 0.0 KN/m

Arm = $Water\ table / 3 =$ 0.73 / 3 = 0.24 m Moment = 0.0 * 0.24 = 0.0 KN-m/m

Wall selfweight seismic effect not considered in calculations

Wall seismic = 0 KN

Moment = 0 KN-m/m

Hor. resultant Rh = 164.8 + 1.8 + 0.0 + 0.0 + 6.5 + 0.0 + 0.0 + 0.0 = 173.1 KN/m

Overturning moment OTM = 426.0 + 0.4 + 0.0 + 0.0 + 29.8 + 0.0 + 0.0 + 0.0 = 456.2 KN-m/m

Arm of hor. resultant = $OTM / Rh =$ 456.2 / 173.1 = 2.63 m

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- Resisting Forces

Stem weight *WStem = DL factor * Thickness * Height * γc* =1.0 * 43.2 / 100 * 7.50 * 24.00 = 77.8 KN/m

Arm = *Toe + ΔThickness / 2* = 1.06 + (68.7 - 43.2) / 100 / 2 = 1.53 m Moment = 77.8 * 1.53 = 119.1 KN-m/m

Stem taper *WTaper = DL factor * ΔThick * Height / 2 * γc* = 1.0 * (68.7 - 43.2) / 100 * 7.50 / 2 * 24.00 = 23.0 KN/m

Arm = *Toe + ΔThick * 2 / 3* = 1.06 + (68.7 + 43.2) / 100 / 3 = 1.23 m

Moment = 23.0 * 1.23 = 28.2 KN-m/m

CMU stem at top = 0.0 KN/m

Arm = *Toe + Thickness / 2* = 1.06 + 0.0 / 100 / 2 = 0.00 m

Moment = 0.0 * 0.00 = 0.0 KN-m/m

Ftg. weight *WFtg = DL factor * Length * Thickness * γc* =1.0 * 3.75 * 70.0 / 100 * 24.00 = 62.9 KN/m

Arm = *Length / 2* = 3.75 / 2 = 1.87 m Moment = 62.9 * 1.87 = 117.9 KN-m/m

Key weight *WKey = DL factor * Depth * Thickness * γc* =1.0 * 0.00 / 100 * 0.0 / 100 * 24.00 = 0.0 KN/m

Arm = *Toe + Thickness / 2* = 1.06 + 0.0 / 100 / 2 = 1.06 m Moment = 0.0 * 1.06 = 0.0 KN-m/m

Soil cover = *DL factor * Toe * Soil cover * γ* =1.0 * 1.06 * 0.30 * 19.0 = 6.0 KN/m

Arm = *Toe / 2* = 1.06 / 2 = 0.53 m Moment = 6.0 * 0.53 = 3.2 KN-m/m

Stem wedge = *DL factor * ΔThick * Height / 2 * γ* = 1.0 * (68.7 - 43.2) / 100 * 6.70 / 2 * 19.0 = 16.2 KN/m

Arm = *Toe + ΔThick * 2 / 3* = 1.06 + (68.7 - 43.2) / 100 * 2 / 3 = 1.23 m

Moment = 16.2 * 1.23 = 20.0 KN-m/m

Backfill weight = *DL factor * Heel * Height * γ* =1.0 * 2.00 * 6.70 * 19.0 = 254.6 KN/m

Arm = *Ftg - Heel / 2* = 3.75 - 2.00 / 2 = 2.75 m Moment = 254.6 * 2.75 = 699.4 KN-m/m

Backfill slope = *DL factor * Heel * Wedge / 2 * γ* =

= 1.0 * 2.0 * 0.35 / 2 * 19.0 = 6.7 KN/m

Arm = *ftg - Heel / 3* = 3.75 - 2.00 / 3 = 3.08 m

Moment = 6.7 * 3.08 = 20.6 KN-m/m

Water = *DL factor * Heel * (Water - Ftg) * (γs - γ)* = 1.0 * 2.00 * (0.73 - 70.0 / 100) * (19.0 - 19.0) = 0.0 KN/m

Arm = *Ftg - Heel / 2* = 3.75 - 2.00 / 2 = 2.75 m Moment = 0.0 * 2.75 = 0.0 KN-m/m

Seismic Pae-Pa = *EQ factor * (Paev - Pav)* = 0.0 * (50.2 - 48.8) = 0.0 KN/m

Arm = *Footing length* = 3.75 m Moment = 0.0 * 3.75 = 0.0 KN-m/m

Backfill Pav = *Lat factor * Pav* = 1.0 * 48.8 = 48.8 KN/m

Arm = *Footing length* = 3.75 m Moment = 48.8 * 3.75 = 182.9 KN-m/m

Concentrated = *DL factor * Ver load + LL factor * Ver load* = 1.0 * 0.0 + 0.0 * 0.0 = 0.0 KN/m

Arm = *Toe + Stem - Ecc* = 1.06 + (68.7 - 0.0) / 100 = 1.75 m

Moment = 0.0 * 1.75 = 0.0 KN-m/m

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Surcharge = *Srch factor * Heel * Surcharge* = 1.0 * 2.0 * 0.0 = 0.0 KN/m

Arm = *ftg - Heel / 2* = 3.75 - 2.00 / 2 = 2.75 m

Moment = 0.0 * 2.75 = 0.0 KN-m/m

Strip = *Strip factor * Surcharge * Heel* = 1.0 * 0.0 * 2.00 = 0.0 KN/m

Arm = *Footing - Heel / 2* = 3.75 - 2.00 / 2 = 2.75 m Moment = 0.0 * 2.75 = 0.0 KN-m/m

Buoyancy = *DL factor * yw * Water table * Ftg* = 1.0 * 9.8 * 0.73 * 3.75 = -26.8 KN/m

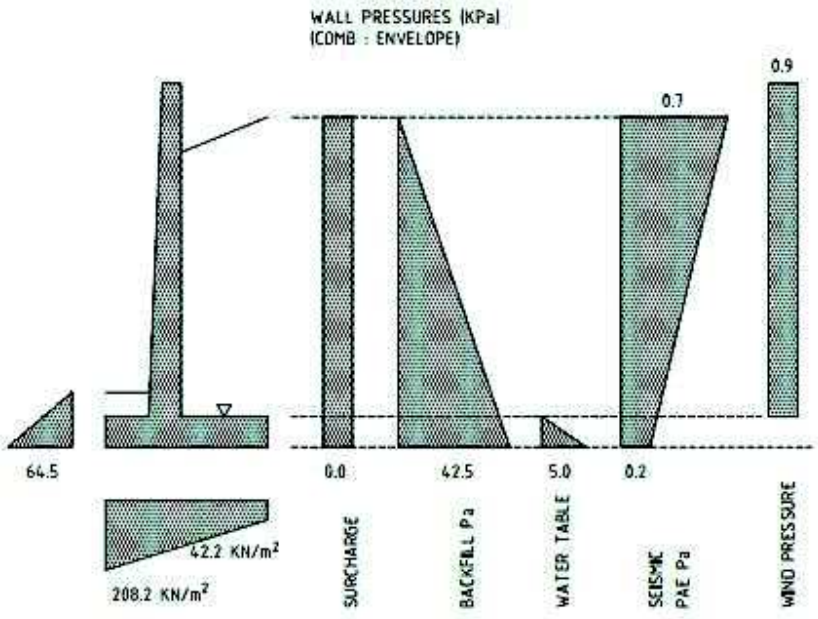
Arm = *Footing / 2* = 3.75 / 2 = 1.87 m Moment = -26.8 * 1.87 = -50.3 KN-m/m

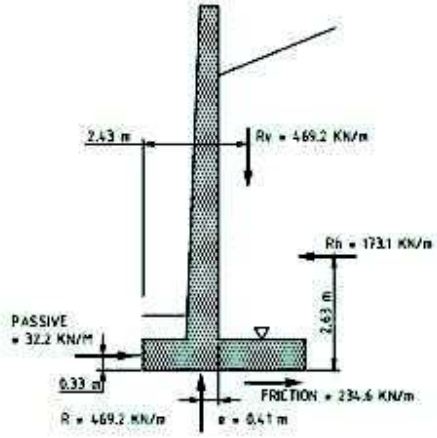
Ver. resultant Rv = *Σ Vertical forces* = 469.2 KN/m

Resisting moment RM = *Σ Moments* = 1141.1 KN-m/m

Arm of ver. resultant = *RM / Rv* = 1141.1 / 469.2 = 2.43 m

Overturning ratio = *RM / OTM* = 1141.1 / 456.2 = 2.50 > 1.50 OK





SOIL BEARING PRESSURES (Comb. D+H+W)

$$\text{Eccentricity} = \frac{Ftg}{2} - \frac{RM - OTM}{Rv} = \frac{3.75}{2} - \frac{1141.1 - 456.2}{469.2} = 0.41 \text{ m}$$

Bearing length = *Min (Ftg , 3 * (Ftg / 2 - Ecc))* = Min (3.75, 3 * (3.75 / 2 - 0.41)) = 3.75 m

$$\text{Toe bearing} = \frac{Rv}{Ftg} + \frac{6 * Rv * Ecc}{Ftg^2} = \frac{469.2}{3.75} + \frac{6 * 469.2 * 0.41}{3.75^2} = 208.2 \text{ KN/m}^2 < 600.0 \text{ KN/m}^2 \text{ OK}$$

$$\text{Heel bearing} = \frac{Rv}{Ftg} - \frac{6 * Rv * Ecc}{Ftg^2} = \frac{469.2}{3.75} - \frac{6 * 469.2 * 0.41}{3.75^2} = 42.2 \text{ KN/m}^2$$

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SLIDING CALCULATIONS (Comb. D+H+W)

Passive coefficient *kp = 1 / ka* = 1 / 0.29 = 3.39 KN/m²

Passive depth *Dp = Soil cover + Ftg + Key - Neglect depth* = 0.30 + (70.0 + 0.0) / 100 - 0.00 = 1.00 m

Passive pressure top = *kp * γ * Neglect depth* = 3.39 * 19.0 * 0.00 = 0.00 KN/m²

Passive pressure bot = *kp * γ * (Dp + Neglect depth)* = 3.39 * 19.0 * (1.00 + 0.00) = 64.45 KN/m²

Passive force = *(Pressure top + Pressure bot) / 2 * Dp* = (0.00 + 64.45) / 2 * 1.00 = 32.2 KN/m

Friction force = *Max (0, Rv * Friction coeff.)* = Max (0, 469.2 * 0.50) = 234.6 KN/m

Sliding ratio = *(Passive + Friction) / Rh* = (32.2 + 234.6) / 173.1 = 1.54 > 1.50 OK

STEM DESIGN (Comb. 0.9D+1.6H+1.6W)

Backfill = *Lat factor * Pah* = 1.6 * 122.3 = 197.0 KN/m

Arm = *Hb / 3* = 6.70 / 3 = 2.23 m Moment = 197.0 * 2.23 = 439.9 KN-m/m

Water table = *Lat factor * Pw* = 1.6 * 0.0 = 0.0 KN/m

Arm = *(Water table - Ftg) / 3* = (0.73 - 70.0 / 100) / 3 = 0.01 m Moment = 0.0 * 0.01 = 0.0 KN-m/m

Surcharge = *Lat factor * ka * Surcharge * Hb* = 1.6 * 0.31 * 0.0 * 6.70 = 0.0 KN/m

Arm = *Hb / 2* = 6.70 / 2 = 3.35 m Moment = 0.0 * 3.35 = 0.0 KN-m/m

Strip load = *Σ Lat factor * 2 * Q / π * [β - Sin β * Cos (2α)]* = 0.0 KN/m

Arm = 3.35 m Moment = 0.0 * 3.35 = 0.0 KN-m/m

Wind load = *WL factor * Pressure * Wind height* = 1.6 * 0.9 * 7.20 = 10.4 KN/m

Arm = *Stem - Wind height / 2* = 7.50 - 7.20 / 2 = 3.90 m Moment = 10.4 * 3.90 = 40.4 KN-m/m

Backfill seismic = *EQ factor * (Paeh - Pah)* = 0.0 * (126.6 - 123.1) = 0.0 KN/m

Arm = *0.6 * Hb* = 0.6 * 6.70 = 4.02 m Moment = 0.0 * 4.02 = 0.0 KN-m/m

Water seismic = *EQ factor * Pwe* = 0.0 * 0.0 = 0.0 KN/m

Arm = *(Water table - Ftg) / 3* = (0.73 - 70.0 / 100) / 3 = 0.01 m Moment = 0.0 * 0.01 = 0.0 KN-m/m

Max. shear = 197.0 + 0.0 + 0.0 + 0.0 + 10.4 + 0.0 + 0.0 = 207.3 KN/m

Shear at critical section = *Max shear - Max shear / Hb * d* = 207.3 - 207.3 / 6.70 * 60.3 / 100 = 188.7 KN/m

Max. moment = 439.9 + 0.0 + 0.0 + 0.0 + 40.4 + 0.0 + 0.0 = 480.3 KN-m/m

Shear strength *φ Vn = φ * 0.17 * (f'c)½ * 10 * d* ACI Eq. (11-3)

φ Vn = 0.75 * 0.17 * (35)½ * 10 * 60.3 = 444.3 KN/m > 188.7 KN/m OK

Use M25 @ 20.0 cm As = 25.00 cm²/m *ρ = As / b d* = 25.00 / (100 * 62.4) = 0.0040

Bending strength *φ Mn = φ * d² * f'c * q * (1 - 0.59 * q)* ACI 10.2.7

φ Mn = 0.90 * 62.4² * 35.0 * 0.048 * (1 - 0.59 * 0.048) = 573.3 KN-m/m > 480.3 KN-m/m OK

Hooked *Ldh = 0.24 * fy / (f'c)½ * db * 0.7* = 0.24 * 420.0 / (35.0)½ * 2.52 * 0.7 * 0.84 = 25.2 cm ACI 12.5

Dev. length at footing = *Ftg - Cover* = 70.0 - 5.0 = 65.0 cm > 25.2 cm OK

*As min = Min (Max ((1.4 / fy, 0.25 * √f'c / fy) * b * d), 1.33 * As req)* ACI 10.5

= Min (Max ((1.4 / 420, 0.25 * (35.0)½ / 420) * 100 * 62.4, 1.33 * 19.7) = 22.0 cm²/m

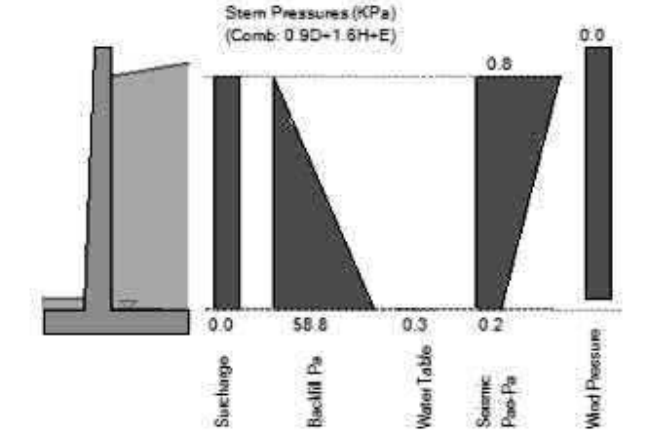
*As S&T = 0.0018 * b * t* = 0.0018 * 100.0 * 68.7 = 12.37 cm²/m ACI 7.12.2.1

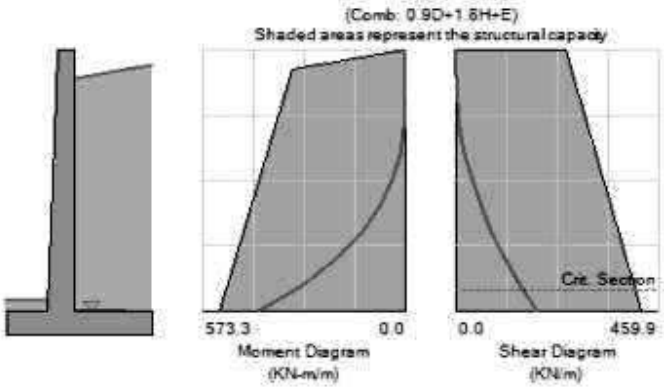
Min. steel area ratio = *As min / As* = 21.99 / 25.00 = 0.88 < 1.0 OK

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HEEL DESIGN (Comb. 1.4D)

Bearing force = 0.0 KN/m (Neglect bearing pressure for heel design)

Arm = *(Bearing1 * Heel² / 2 + (Bearing2 - Bearing1) * Heel² / 6) / Force*

= (259.1 * 2.00² / 2 + (150.2 - 259.1) * 2.00² / 6) / 0.0 = 1.09 m

Moment = 0.0 * 1.09 = 0.0 KN-m/m

Concrete weight = *DL factor * Thick * Heel * γc* = 1.4 * 70.0 / 100 * 2.00 * 24.00 = 47.0 KN/m

Arm = *Heel / 2* = 2.00 / 2 = 1.00 m Moment = 47.0 * 1.00 = 47.0 KN-m/m

Backfill weight = *DL factor * Heel * Height * γ* = 1.4 * 2.00 * 6.70 * 19.0 = 356.4 KN/m

Arm = *Heel / 2* = 2.00 / 2 = 1.00 m Moment = 356.4 * 1.00 = 356.4 KN-m/m

Backfill slope = *DL factor * Heel * Wedge / 2 * γ* =

= 1.4 * 2.0 * 0.35 / 2 * 19.0 = 9.4 KN/m

Arm = *Heel * 2 / 3* = 2.00 * 2 / 3 = 1.33 m Moment = 9.4 * 1.33 = 12.5 KN-m/m

Water = *DL factor * Heel * Water table * (γs - γ)* = 1.4 * 2.00 * 0.73 * (19.0 - 19.0) = 0.0 KN/m

Arm = *Heel / 2* = 2.00 / 2 = 1.00 m Moment = 0.0 * 1.00 = 0.0 KN-m/m

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Surcharge = *Srch factor * Heel * Surcharge* = 1.4 * 2.0 * 0.0 = 0.0 KN/m

Arm = *Heel / 2* = 2.00 / 2 = 1.00 m Moment = 0.0 * 1.00 = 0.0 KN-m/m

Strip = *Strip factor * Surcharge * Width* = 1.4 * 0.0 * 0.00 = 0.0 KN/m

Arm = *Distance + Width / 2* = 0.00 + 0.00 / 2 = 1.00 m

Moment = 0.0 * 1.00 = 0.0 KN-m/m

Max. Shear Vu = -0.0 + 47.0 + 356.4 + 9.4 + 0.0 + 0.0 + 0.0 = 412.9 KN/m

Max. Moment Mu = -0.0 + 47.0 + 356.4 + 12.5 + 0.0 + 0.0 + 0.0 = 416.0 KN/m

Shear strength $\varphi Vn = \varphi * 0.17 * (f'c)^{1/2} * 10 * d$ ACI Eq. (11-3)

φ Vn = 0.75 * 0.17 * (35)½ * 10 * 63.8 = 469.6 KN/m > Vu = 412.9 KN/m OK

Use D25 @ 15.0 cm As = 32.73 cm²/m $\rho = As / b\,d$ = 32.73 / (100 * 63.8) = 0.0051

Bending strength $\varphi Mn = \varphi * d^2 * f'c * q * (1 - 0.59 * q)$ ACI 10.2.7

φ Mn = 0.90 * 63.8² * 35.0 * 0.062 * (1 - 0.59 * 0.062) = 760.1 KN-m/m > Mu = 416.0 KN-m/m OK

Cover factor = *Min (2.5, (Cover + db /2, Spacing / 2) / db)* = Min (2.5, (5.0 + 2.50 / 2, 15.0 / 2) / 2.50) = 2.5

Straight *Ld = Max (30, fy / 1.1 / (f'c)½ * Size * Location / Cover * db * Ratio)* ACI Eq. (12-1)

= Max (30, 420.0 / 1.1 / (35)½ * 1.0 * 1.3 / 2.5 * 2.50 * 0.55) = 83.9 cm

Hooked *Ldh = 0.24 * fy / (f'c)½ * db * 0.7* = 0.24 * 420.0 / (35.0)½ * 2.50 * 0.7 * 0.55 = 29.8 cm ACI 12.5

Dev. length at toe side = *Ftg - Heel - Cover* = (3.75 - 2.00) * 100 - 5.0 = 169.7 cm > 83.9 cm OK

Dev. length at heel side = *Heel - Cover* = 2.00 * 100 - 5.0 = 195.0 cm > 83.9 cm OK

*As min = Min (Max ((1.4 / fy, 0.25 * √f'c / fy) * b * d), 1.33 * As req)* ACI 10.5

= Min (Max ((1.4 / 420, 0.25 * (35.0)½ / 420) * 100 * 63.8, 1.33 * 11.2) = 22.4 cm²/m

*As S&T = 0.0018 * b * t* = 0.0018 * 100.0 * 70.0 = 12.60 cm²/m ACI 7.12.2.1

Min. steel area ratio = *As min / As* = 22.45 / 32.73 = 0.69 < 1.0 OK

TOE DESIGN (Comb. 0.9D+1.6H+1.6W)

Bearing force = *(Bearing1 + Bearing2) / 2 * Toe* = (326.8 + 202.8) / 2 * 1.06 = 280.7 KN/m

Arm = *(Bearing1 * Toe² / 2 + (Bearing2 - Bearing1) * Toe² / 3) / Force*

= (202.8 * 1.06² / 2 + (326.8 - 202.8) * 1.06² / 3) / 280.7 = 0.57 m

Moment = 280.7 * 0.57 = 160.4 KN-m/m

Water Buoyancy = *DL factor * γw * Water table * Toe* = 0.9 * 9.8 * 0.73 * 1.06 = 6.8 KN/m

Arm = *Toe / 2* = 1.06 / 2 = 0.53 m Moment = 6.8 * 0.53 = 3.6 KN-m/m

Concrete weight = *DL factor * Thick * Toe * γc* = 0.9 * 70.0 / 100 * 1.06 * 24.00 = 16.0 KN/m

Arm = *Toe / 2* = 1.06 / 2 = 0.53 m Moment = 16.0 * 0.53 = 8.5 KN-m/m

Soil cover = *DL factor * Toe * Height * γ* = 0.9 * 1.06 * 5.44 * 19.0 = 5.4 KN/m

Arm = *Toe / 2* = 1.06 / 2 = 0.53 m Moment = 5.4 * 0.53 = 2.9 KN-m/m

Max. Shear Vu = 280.7 + 6.8 - 16.0 - 5.4 = 266.0 KN/m

Shear at crit. section Vu = *Max shear * (Toe - d) / Toe* = 266.0 * (1.06 - 63.8 / 100) / 1.06 = 106.0 KN/m

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Max. Moment Mu = 160.4 + 3.6 - 8.5 - 2.9 = 152.6 KN/m

Shear strength $\varphi Vn = \varphi * 0.17 * (f'c)^{1/2} * 10 * d$ ACI Eq. (11-3)

φ Vn = 0.75 * 0.17 * (35)½ * 10 * 63.8 = 469.6 KN/m > Vu = 106.0 KN/m OK

Use D25 @ 15.0 cm As = 32.73 cm²/m $\rho = As / b\,d$ = 32.73 / (100 * 63.8) = 0.0051

Bending strength $\varphi Mn = \varphi * d^2 * f'c * q * (1 - 0.59 * q)$ ACI 10.2.7

φ Mn = 0.90 * 63.8² * 35.0 * 0.062 * (1 - 0.59 * 0.062) = 760.1 KN-m/m > Mu = 152.6 KN-m/m OK

Cover factor = $Min\,(2.5, (Cover + db/2, Spacing/2)/db)$ = Min (2.5, (5.0 + 2.50 / 2, 15.0 / 2) / 2.50) = 2.5

Straight $Ld = Max\,(30, fy / 1.1 / (f'c)^{1/2} * Size * Location / Cover * db * Ratio$ ACI Eq. (12-1)

= Max (30, 420.0 / 1.1 / (35)½ * 1.0 * 1.0 / 2.5 * 2.50 * 0.20) = 64.5 cm

Hooked $Ldh = 0.24 * fy / (f'c)^{1/2} * db * 0.7$ = 0.24 * 420.0 / (35.0)½ * 2.50 * 0.7 * 0.20 = 29.8 cm ACI 12.5

Dev. length at heel side = $Ftg - Toe - Cover$ = (3.75 - 1.06) * 100 - 5.0 = 263.7 cm> 64.5 cm OK

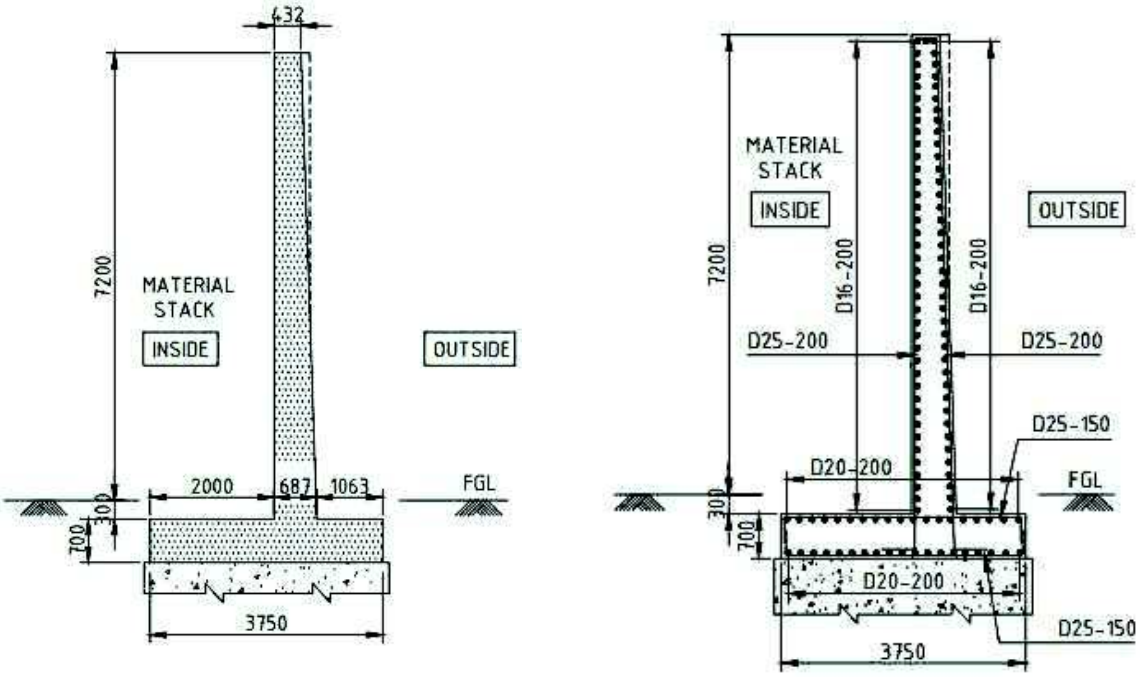
Dev. length at toe side = $Toe - Cover$ = 1.06 * 100 - 5.0 = 101.0 cm> 64.5 cm OK

$As\ min = Min\,(Max\,((1.4 / fy, 0.25 * \sqrt{f'c} / fy) * b * d), 1.33 * As\ req)$ ACI 10.5

= Min (Max ((1.4 / 420, 0.25 * (35.0)½ / 420) * 100 * 63.8, 1.33 * 6.0) = 8.5 cm²/m

$As\ S\&T = 0.0018 * b * t$ = 0.0018 * 100.0 * 70.0 = 12.60 cm²/m ACI 7.12.2.1

Min. steel area ratio = $As\ min / As$ = 12.60 / 32.73 = 0.38 < 1.0 OK



RETANING WALL TYPE - I

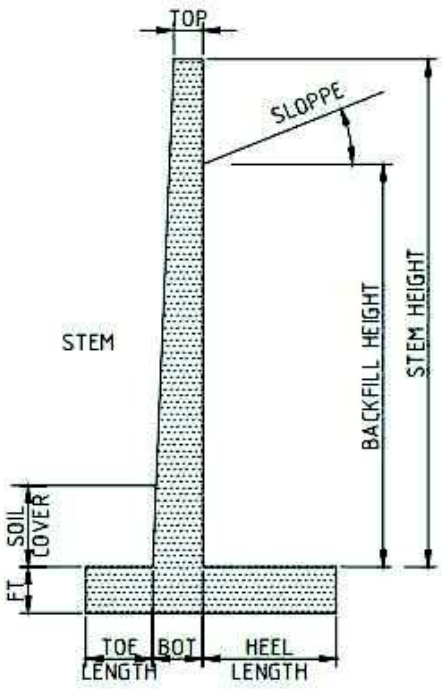
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8. DESIGN OF RETAINING WALL TYPE-II:

Conc. Stem Height	9.35	m	
Stem Thickness Top	43.2	cm	
Stem Thickness Bot	75.0	cm	
Footing Thickness	70.0	cm	
Toe Length	1.00	m	
Heel Length	2.00	m	
Soil Cover @ Toe	2.15	m	
Backfill Height	6.70	m	OK
Backfill Slope Angle	10.0	deg	
Wind Load on Stem	0.9	KPa	
Wind Height from Top	7.20	m	



Total Base width = (2.0+0.687+1.063) = 3.75m; Height = (6.7+0.7)= 7.4m;
Ratio = (3.75 / 7.4) = 0.51; OK [as, 0.5< 0.51 < 0.7]

BACKFILL PROPERTIES

Wall taper α = 0.000 rad

Backfill slope $\beta = slope * \pi / 180$ = 10.0 * 3.14 / 180 = 0.175 rad

Internal friction $\phi = Int.\ friction * \pi / 180$ = 33.0 * 3.14 / 180 = 0.576 rad

Wall-soil friction $\delta = \phi / 2$ = 0.576 / 2 = 0.288 rad

Seismic angle $\theta = aTan\ (kh / (1 - kv))$ = aTan (0.02 / (1 - 0.02)) = 0.020 rad

Footing length $ftg = toe + stem + heel$ = 1.00 + 75.0 / 100 + 2.00 = 3.75 m

Height for Stability $Hs = wedge + backfill + footing$ = 0.35 + 6.70 + 70.0 / 100 = 7.75 m

Earth pressure theory = Coulomb Active Moist density = 19 KN/m³ Saturated density = 19 KN/m³

Active coefficient $ka = \dfrac{Cos^2(\phi + \alpha)}{Cos^2\alpha * Cos(\delta + \alpha) * [1 + (\ \frac{Sin(\phi + \delta) * Sin(\phi - \beta)}{Cos(\delta + \alpha) * Cos(\beta - \alpha)}\ )^{1/2}\ ]^2}$ = 0.30

Active pressure $pa = ka * \gamma$ = 0.30 * 19.0 = 5.7 KPa/m of height

- For stability analysis (non-seismic)

Active force $Pa = ka * \gamma * Hs^2 / 2$ = 0.30 * 19.0 * 7.75² / 2 = 171.9 KN/m

$Pah = Pa * Cos(\delta + \alpha)$ = 171.9 * Cos(0.288 + 0.000) = 164.8 KN/m

$Pav = Pa * Sin(\delta + \alpha)$ = 171.9 * Sin(0.288 + 0.000) = 48.8 KN/m

Water force $Pw = (ka * (\gamma s - \gamma w - \gamma) + \gamma w) * (Water\ table)^2 / 2$

Pw = (0.30 * (19.0 - 9.8 - 19.0) + 9.8) * 0.73² / 2 = 1.8 KN/m

- For stem design (non-seismic)

Active force $Pa = ka * \gamma * H^2 / 2$ = 0.30 * 19.0 * 6.70² / 2 = 128.4 KN/m

$Pah = Pa * Cos(\delta + \alpha)$ = 128.4 * Cos (0.288 + 0.000) = 123.1 KN/m

$Pav = Pa * Sin(\delta + \alpha)$ = 128.4 * Sin (0.288 + 0.000) = 36.5 KN/m

Water force $Pw = (ka * (\gamma s - \gamma w - \gamma) + \gamma w) * (Water\ table - Ftg)^2 / 2$

Pw = (0.30 * (19.0 - 9.8 - 19.0) + 9.8) * (0.73 - 70.0 / 100)² / 2 = 0.0 KN/m

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Hor. seismic coeff. kh = 0.02 Ver. seismic coeff. kv = 0.02

Active seismic coeff. $kae = \frac{Cos^2(\phi - \alpha - \theta)}{Cos\theta * Cos^2\alpha * Cos(\delta + \alpha + \theta) * [1 + (\frac{Sin(\phi + \delta) * Sin(\phi - \beta - \theta)}{Cos(\delta + \alpha + \theta) * Cos(\beta - \alpha)})^{½}]^2} = 0.32$

- For stability analysis (seismic)

Seismic force $Pae = kae * \gamma * Hs^2 / 2 * (1 - kv) =$ 0.32 * 19.0 * 7.75² / 2 * (1 - 0.0) = 176.7 KN/m

$Paeh = Pae * Cos(\delta + \alpha) =$ 176.7 * Cos (0.288 + 0.000) = 169.4 KN/m

$Paev = Pae * Sin(\delta + \alpha) =$ 176.7 * Sin (0.288 + 0.000) = 50.2 KN/m

Water force $Pwe = kh * (\gamma s - \gamma) * (water)^2 / 2$

Pwe = 0.02 * (19.0 - 19.0) * (0.73)² / 2 = 0.0 KN/m

- For stem design (seismic)

Seismic force $Pae = kae * \gamma * H^2 / 2 =$ 0.32 * 19.0 * 6.70² / 2 = 132.0 KN/m

$Paeh = Pae * Cos(\delta + \alpha) =$ 132.0 * Cos (0.288 + 0.000) = 126.6 KN/m

$Paev = Pae * Sin(\delta + \alpha) =$ 132.0 * Sin (0.288 + 0.000) = 37.5 KN/m

Water force $Pwe = kh * (\gamma s - \gamma) * (water\ table - ftg)^2 / 2$

Pwe = 0.02 * (19.0 - 19.0) * (0.73 - 70.0 / 100)² / 2 = 0.0 KN/m

OVERTURNING CALCULATIONS (Comb. D+H+W)

- Overturning Forces

Backfill = $Lat\ factor * Pah =$ 1.0 * 164.8 = 164.8 KN/m

Arm = $Hs / 3 =$ 7.75 / 3 = 2.58 m Moment = 164.8 * 2.58 = 426.0 KN-m/m

Water table = $Lat\ factor * Pw =$ 1.0 * 1.8 = 1.8 KN/m

Arm = $Water\ table / 3 =$ 0.73 / 3 = 0.24 m Moment = 1.8 * 0.24 = 0.4 KN-m/m

Surcharge = $Lat\ factor * ka * Surcharge * Hs =$ 1.0 * 0.31 * 0.0 * 7.75 = 0.0 KN/m

Arm = $Hs / 2 =$ 7.75 / 2 = 3.88 m Moment = 0.0 * 3.88 = 0.0 KN-m/m

Strip load = $\Sigma\ Lat\ factor * 2 * Q / \pi * [\beta - Sin\beta * Cos(2\alpha)] =$ 0.0 KN/m

Arm = 3.35 m Moment = 0.0 * 3.35 = 0.0 KN-m/m

Wind load = $WL\ factor * Pressure * Wind\ height =$ 1.0 * 0.9 * 7.20 = 6.5 KN/m

Arm = $Ftg + Stem - Wind\ height / 2 =$ 70.0 / 100 + 9.35 - 7.20 / 2 = 6.45 m

Moment = 6.5 * 6.45 = 41.8 KN-m/m

Backfill seismic = $EQ\ factor * (Paeh - Pah) =$ 0.0 * (169.4 - 164.8) = 0.0 KN/m

Arm = $0.6 * Hs =$ 0.6 * 7.75 = 4.65 m Moment = 0.0 * 4.65 = 0.0 KN-m/m

Water seismic = $EQ\ factor * Pwe =$ 0.0 * 0.0 = 0.0 KN/m

Arm = $Water\ table / 3 =$ 0.73 / 3 = 0.24 m Moment = 0.0 * 0.24 = 0.0 KN-m/m

Wall selfweight seismic effect not considered in calculations

Wall seismic = 0 KN

Moment = 0 KN-m/m

Hor. resultant Rh = 164.8 + 1.8 + 0.0 + 0.0 + 6.5 + 0.0 + 0.0 + 0.0 = 173.1 KN/m

Overturning moment OTM = 426.0 + 0.4 + 0.0 + 0.0 + 41.8 + 0.0 + 0.0 + 0.0 = 468.2 KN-m/m

Arm of hor. resultant = $OTM / Rh =$ 468.2 / 173.1 = 2.70 m

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- Resisting Forces

Stem weight *WStem = DL factor * Thickness * Height * γc* =1.0 * 43.2 / 100 * 9.35 * 24.00 = 96.9 KN/m

Arm = *Toe + ΔThickness / 2* = 1.00 + (75.0 - 43.2) / 100 / 2 = 1.53 m Moment = 96.9 * 1.53 = 148.7 KN-m/m

Stem taper *WTaper = DL factor * ΔThick * Height / 2 * γc* = 1.0 * (75.0 - 43.2) / 100 * 9.35 / 2 * 24.00 = 35.7 KN/m

Arm = *Toe + ΔThick * 2 / 3* = 1.00 + (75.0 + 43.2) / 100 / 3 = 1.21 m

Moment = 35.7 * 1.21 = 43.2 KN-m/m

CMU stem at top = 0.0 KN/m

Arm = *Toe + Thickness / 2* = 1.00 + 0.0 / 100 / 2 = 0.00 m

Moment = 0.0 * 0.00 = 0.0 KN-m/m

Ftg. weight *WFtg = DL factor * Length * Thickness * γc* =1.0 * 3.75 * 70.0 / 100 * 24.00 = 63.0 KN/m

Arm = *Length / 2* = 3.75 / 2 = 1.88 m Moment = 63.0 * 1.88 = 118.1 KN-m/m

Key weight *WKey = DL factor * Depth * Thickness * γc* =1.0 * 0.00 / 100 * 0.0 / 100 * 24.00 = 0.0 KN/m

Arm = *Toe + Thickness / 2* = 1.00 + 0.0 / 100 / 2 = 1.00 m Moment = 0.0 * 1.00 = 0.0 KN-m/m

Soil cover = *DL factor * Toe * Soil cover * γ* =1.0 * 1.00 * 2.15 * 19.0 = 40.9 KN/m

Arm = *Toe / 2* = 1.00 / 2 = 0.50 m Moment = 40.9 * 0.50 = 20.4 KN-m/m

Stem wedge = *DL factor * ΔThick * Height / 2 * γ* = 1.0 * (75.0 - 43.2) / 100 * 6.70 / 2 * 19.0 = 20.2 KN/m

Arm = *Toe + ΔThick * 2 / 3* = 1.00 + (75.0 - 43.2) / 100 * 2 / 3 = 1.21 m

Moment = 20.2 * 1.21 = 24.5 KN-m/m

Backfill weight = *DL factor * Heel * Height * γ* =1.0 * 2.00 * 6.70 * 19.0 = 254.6 KN/m

Arm = *Ftg - Heel / 2* = 3.75 - 2.00 / 2 = 2.75 m Moment = 254.6 * 2.75 = 700.2 KN-m/m

Backfill slope = *DL factor * Heel * Wedge / 2 * γ* =

= 1.0 * 2.0 * 0.35 / 2 * 19.0 = 6.7 KN/m

Arm = *ftg - Heel / 3* = 3.75 - 2.00 / 3 = 3.08 m

Moment = 6.7 * 3.08 = 20.7 KN-m/m

Water = *DL factor * Heel * (Water - Ftg) * (γs - γ)* = 1.0 * 2.00 * (0.73 - 70.0 / 100) * (19.0 - 19.0) = 0.0 KN/m

Arm = *Ftg - Heel / 2* = 3.75 - 2.00 / 2 = 2.75 m Moment = 0.0 * 2.75 = 0.0 KN-m/m

Seismic Pae-Pa = *EQ factor * (Paev - Pav)* = 0.0 * (50.2 - 48.8) = 0.0 KN/m

Arm = *Footing length* = 3.75 m Moment = 0.0 * 3.75 = 0.0 KN-m/m

Backfill Pav = *Lat factor * Pav* = 1.0 * 48.8 = 48.8 KN/m

Arm = *Footing length* = 3.75 m Moment = 48.8 * 3.75 = 183.1 KN-m/m

Concentrated = *DL factor * Ver load + LL factor * Ver load* = 1.0 * 0.0 + 0.0 * 0.0 = 0.0 KN/m

Arm = *Toe + Stem - Ecc* = 1.00 + (75.0 - 0.0) / 100 = 1.75 m

Moment = 0.0 * 1.75 = 0.0 KN-m/m

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Surcharge = *Srch factor * Heel * Surcharge* = 1.0 * 2.0 * 0.0 = 0.0 KN/m

Arm = *ftg - Heel / 2* = 3.75 - 2.00 / 2 = 2.75 m

Moment = 0.0 * 2.75 = 0.0 KN-m/m

Strip = *Strip factor * Surcharge * Heel* = 1.0 * 0.0 * 2.00 = 0.0 KN/m

Arm = *Footing - Heel / 2* = 3.75 - 2.00 / 2 = 2.75 m Moment = 0.0 * 2.75 = 0.0 KN-m/m

Buoyancy = *DL factor * yw * Water table * Ftg* = 1.0 * 9.8 * 0.73 * 3.75 = -26.9 KN/m

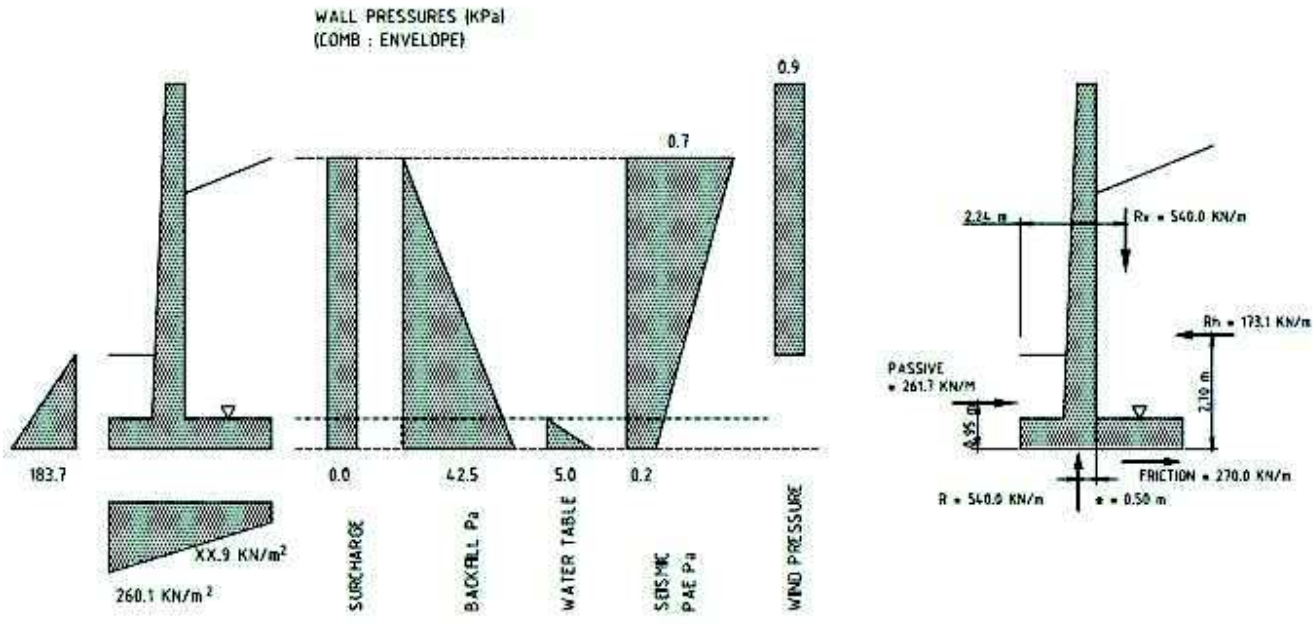
Arm = *Footing / 2* = 3.75 / 2 = 1.88 m Moment = -26.9 * 1.88 = -50.4 KN-m/m

Ver. resultant Rv = *Σ Vertical forces* = 540.0 KN/m

Resisting moment RM = *Σ Moments* = 1208.6 KN-m/m

Arm of ver. resultant = *RM / Rv* = 1208.6 / 540.0 = 2.24 m

Overturning ratio = *RM / OTM* = 1208.6 / 468.2 = 2.58 > 1.50 OK



Eccentricity = $\frac{Ftg}{2} - \frac{RM - OTM}{Rv} = \frac{3.75}{2} - \frac{1208.6 - 468.2}{540.0}$ = 0.50 m

Bearing length = *Min (Ftg , 3 * (Ftg / 2 - Ecc))* = Min (3.75, 3 * (3.75 / 2 - 0.50)) = 3.75 m

Toe bearing = $\frac{Rv}{Ftg} + \frac{6 * Rv * Ecc}{Ftg^2} = \frac{540.0}{3.75} + \frac{6 * 540.0 * 0.50}{3.75^2}$ = 260.1 KN/m²< 600.0 KN/m² OK

Heel bearing = $\frac{Rv}{Ftg} - \frac{6 * Rv * Ecc}{Ftg^2} = \frac{540.0}{3.75} - \frac{6 * 540.0 * 0.50}{3.75^2}$ = 27.9 KN/m²

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SLIDING CALCULATIONS (Comb. D+H+W)

Passive coefficient *kp = 1 / ka* = 1 / 0.29 = 3.39 KN/m²

Passive depth *Dp = Soil cover + Ftg + Key - Neglect depth* = 2.15 + (70.0 + 0.0) / 100 - 0.00 = 2.85 m

Passive pressure top = *kp * γ * Neglect depth* = 3.39 * 19.0 * 0.00 = 0.00 KN/m²

Passive pressure bot = *kp * γ * (Dp + Neglect depth)* = 3.39 * 19.0 * (2.85 + 0.00) = 183.68 KN/m²

Passive force = *(Pressure top + Pressure bot) / 2 * Dp* = (0.00 + 183.68) / 2 * 2.85 = 261.7 KN/m

Friction force = *Max (0, Rv * Friction coeff.)* = Max (0, 540.0 * 0.50) = 270.0 KN/m

Sliding ratio = *(Passive + Friction) / Rh* = (261.7 + 270.0) / 173.1 = 3.07 > 1.50 OK

STEM DESIGN (Comb. 0.9D+1.6H+1.6W)

Backfill = *Lat factor * Pah* = 1.6 * 122.3 = 197.0 KN/m

Arm = *Hb / 3* = 6.70 / 3 = 2.23 m Moment = 197.0 * 2.23 = 439.9 KN-m/m

Water table = *Lat factor * Pw* = 1.6 * 0.0 = 0.0 KN/m

Arm = *(Water table - Ftg) / 3* = (0.73 - 70.0 / 100) / 3 = 0.01 m Moment = 0.0 * 0.01 = 0.0 KN-m/m

Surcharge = *Lat factor * ka * Surcharge * Hb* = 1.6 * 0.31 * 0.0 * 6.70 = 0.0 KN/m

Arm = *Hb / 2* = 6.70 / 2 = 3.35 m Moment = 0.0 * 3.35 = 0.0 KN-m/m

Strip load = *Σ Lat factor * 2 * Q / π * [β - Sin β * Cos (2α)]* = 0.0 KN/m

Arm = 3.35 m Moment = 0.0 * 3.35 = 0.0 KN-m/m

Wind load = *WL factor * Pressure * Wind height* = 1.6 * 0.9 * 7.20 = 10.4 KN/m

Arm = *Stem - Wind height / 2* = 9.35 - 7.20 / 2 = 5.75 m Moment = 10.4 * 5.75 = 59.6 KN-m/m

Backfill seismic = *EQ factor * (Paeh - Pah)* = 0.0 * (126.6 - 123.1) = 0.0 KN/m

Arm = *0.6 * Hb* = 0.6 * 6.70 = 4.02 m Moment = 0.0 * 4.02 = 0.0 KN-m/m

Water seismic = *EQ factor * Pwe* = 0.0 * 0.0 = 0.0 KN/m

Arm = *(Water table - Ftg) / 3* = (0.73 - 70.0 / 100) / 3 = 0.01 m Moment = 0.0 * 0.01 = 0.0 KN-m/m

Max. shear = 197.0 + 0.0 + 0.0 + 0.0 + 10.4 + 0.0 + 0.0 = 207.3 KN/m

Shear at critical section = *Max shear - Max shear / Hb * d* = 207.3 - 207.3 / 6.70 * 66.4 / 100 = 186.8 KN/m

Max. moment = 439.9 + 0.0 + 0.0 + 0.0 + 59.6 + 0.0 + 0.0 = 499.5 KN-m/m

Shear strength *φ Vn = φ * 0.17 * (f'c)½ * 10 * d* ACI Eq. (11-3)

φ Vn = 0.75 * 0.17 * (35)½ * 10 * 66.4 = 489.1 KN/m > 186.8 KN/m OK

Use M25 @ 20.0 cm As = 25.00 cm²/m *ρ = As / b d* = 25.00 / (100 * 68.7) = 0.0036

Bending strength *φ Mn = φ * d² * f'c * q * (1 - 0.59 * q)* ACI 10.2.7

φ Mn = 0.90 * 68.7² * 35.0 * 0.044 * (1 - 0.59 * 0.044) = 632.9 KN-m/m > 499.5 KN-m/m OK

Hooked *Ldh = 0.24 * fy / (f'c)½ * db * 0.7* = 0.24 * 420.0 / (35.0)½ * 2.52 * 0.7 * 0.79 = 23.7 cm ACI 12.5

Dev. length at footing = *Ftg - Cover* = 70.0 - 5.0 = 65.0 cm > 23.7 cm OK

*As min = Min (Max ((1.4 / fy, 0.25 * √f'c / fy) * b * d), 1.33 * As req)* ACI 10.5

= Min (Max ((1.4 / 420, 0.25 * (35.0)½ / 420) * 100 * 68.7, 1.33 * 17.8) = 24.2 cm²/m

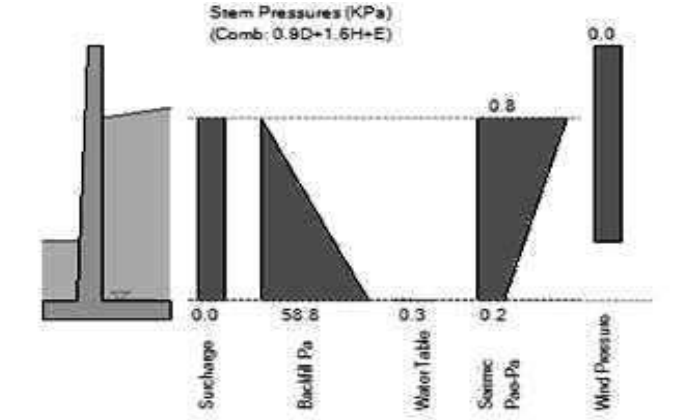
*As S&T = 0.0018 * b * t* = 0.0018 * 100.0 * 75.0 = 13.50 cm²/m ACI 7.12.2.1

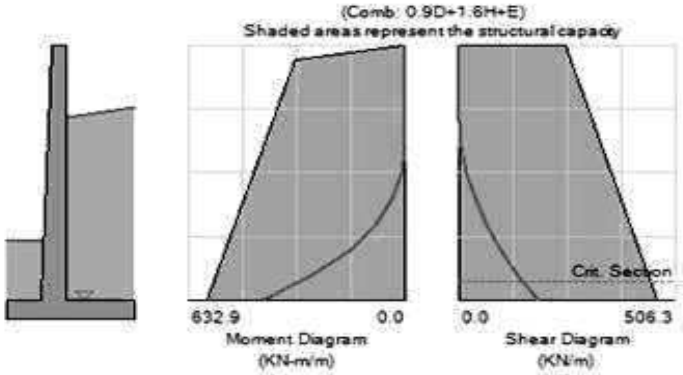
Min. steel area ratio = *As min / As* = 24.21 / 25.00 = 0.97 < 1.0 OK

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HEEL DESIGN (Comb. 1.4D)

Bearing force = 0.0 KN/m (Neglect bearing pressure for heel design)

Arm = *(Bearing1 * Heel² / 2 + (Bearing2 - Bearing1) * Heel² / 6) / Force*

= (245.8 * 2.00² / 2 + (179.2 - 245.8) * 2.00² / 6) / 0.0 = 1.05 m

Moment = 0.0 * 1.05 = 0.0 KN-m/m

Concrete weight = *DL factor * Thick * Heel * γc* = 1.4 * 70.0 / 100 * 2.00 * 24.00 = 47.0 KN/m

Arm = *Heel / 2* = 2.00 / 2 = 1.00 m Moment = 47.0 * 1.00 = 47.0 KN-m/m

Backfill weight = *DL factor * Heel * Height * γ* = 1.4 * 2.00 * 6.70 * 19.0 = 356.4 KN/m

Arm = *Heel / 2* = 2.00 / 2 = 1.00 m Moment = 356.4 * 1.00 = 356.4 KN-m/m

Backfill slope = *DL factor * Heel * Wedge / 2 * γ* =

= 1.4 * 2.0 * 0.35 / 2 * 19.0 = 9.4 KN/m

Arm = *Heel * 2 / 3* = 2.00 * 2 / 3 = 1.33 m Moment = 9.4 * 1.33 = 12.5 KN-m/m

Water = *DL factor * Heel * Water table * (γs - γ)* = 1.4 * 2.00 * 0.73 * (19.0 - 19.0) = 0.0 KN/m

Arm = *Heel / 2* = 2.00 / 2 = 1.00 m Moment = 0.0 * 1.00 = 0.0 KN-m/m

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Surcharge = *Srch factor * Heel * Surcharge* = 1.4 * 2.0 * 0.0 = 0.0 KN/m

Arm = *Heel / 2* = 2.00 / 2 = 1.00 m Moment = 0.0 * 1.00 = 0.0 KN-m/m

Strip = *Strip factor * Surcharge * Width* = 1.4 * 0.0 * 0.00 = 0.0 KN/m

Arm = *Distance + Width / 2* = 0.00 + 0.00 / 2 = 1.00 m

Moment = 0.0 * 1.00 = 0.0 KN-m/m

Max. Shear Vu = -0.0 + 47.0 + 356.4 + 9.4 + 0.0 + 0.0 + 0.0 = 412.9 KN/m

Max. Moment Mu = -0.0 + 47.0 + 356.4 + 12.5 + 0.0 + 0.0 + 0.0 = 416.0 KN/m

Shear strength $\varphi Vn = \varphi * 0.17 * (f'c)^{1/2} * 10 * d$ ACI Eq. (11-3)

φ Vn = 0.75 * 0.17 * (35)½ * 10 * 63.8 = 469.6 KN/m > Vu = 412.9 KN/m OK

Use D25 @ 15.0 cm As = 32.73 cm²/m $\rho = As / b\,d$ = 32.73 / (100 * 63.8) = 0.0051

Bending strength $\varphi Mn = \varphi * d^2 * f'c * q * (1 - 0.59 * q)$ ACI 10.2.7

φ Mn = 0.90 * 63.8² * 35.0 * 0.062 * (1 - 0.59 * 0.062) = 760.1 KN-m/m > Mu = 416.0 KN-m/m OK

Cover factor = *Min (2.5, (Cover + db /2, Spacing / 2) / db)* = Min (2.5, (5.0 + 2.50 / 2, 15.0 / 2) / 2.50) = 2.5

Straight *Ld = Max (30, fy / 1.1 / (f'c)½ * Size * Location / Cover * db * Ratio)* ACI Eq. (12-1)

= Max (30, 420.0 / 1.1 / (35)½ * 1.0 * 1.3 / 2.5 * 2.50 * 0.55) = 83.9 cm

Hooked *Ldh = 0.24 * fy / (f'c)½ * db * 0.7* = 0.24 * 420.0 / (35.0)½ * 2.50 * 0.7 * 0.55 = 29.8 cm ACI 12.5

Dev. length at toe side = *Ftg - Heel - Cover* = (3.75 - 2.00) * 100 - 5.0 = 170.0 cm> 83.9 cm OK

Dev. length at heel side = *Heel - Cover* = 2.00 * 100 - 5.0 = 195.0 cm> 83.9 cm OK

*As min = Min (Max ((1.4 / fy, 0.25 * √f'c / fy) * b * d), 1.33 * As req)* ACI 10.5

= Min (Max ((1.4 / 420, 0.25 * (35.0)½ / 420) * 100 * 63.8, 1.33 * 11.2) = 22.4 cm²/m

*As S&T = 0.0018 * b * t* = 0.0018 * 100.0 * 70.0 = 12.60 cm²/m ACI 7.12.2.1

Min. steel area ratio = *As min / As* = 22.45 / 32.73 = 0.69 < 1.0 OK

TOE DESIGN (Comb. 0.9D+1.6H+1.6W)

Bearing force = *(Bearing1 + Bearing2) / 2 * Toe* = (386.4 + 242.9) / 2 * 1.00 = 314.7 KN/m

Arm = *(Bearing1 * Toe² / 2 + (Bearing2 - Bearing1) * Toe² / 3) / Force*

= (242.9 * 1.00² / 2 + (386.4 - 242.9) * 1.00² / 3) / 314.7 = 0.54 m

Moment = 314.7 * 0.54 = 169.3 KN-m/m

Water Buoyancy = *DL factor * γw * Water table * Toe* = 0.9 * 9.8 * 0.73 * 1.00 = 6.4 KN/m

Arm = *Toe / 2* = 1.00 / 2 = 0.50 m Moment = 6.4 * 0.50 = 3.2 KN-m/m

Concrete weight = *DL factor * Thick * Toe * γc* = 0.9 * 70.0 / 100 * 1.00 * 24.00 = 15.1 KN/m

Arm = *Toe / 2* = 1.00 / 2 = 0.50 m Moment = 15.1 * 0.50 = 7.6 KN-m/m

Soil cover = *DL factor * Toe * Height * γ* = 0.9 * 1.00 * 36.77 * 19.0 = 36.8 KN/m

Arm = *Toe / 2* = 1.00 / 2 = 0.50 m Moment = 36.8 * 0.50 = 18.4 KN-m/m

Max. Shear Vu = 314.7 + 6.4 - 15.1 - 36.8 = 269.2 KN/m

Shear at crit. section Vu = *Max shear * (Toe - d) / Toe* = 269.2 * (1.00 - 63.8 / 100) / 1.00 = 97.6 KN/m

Max. Moment Mu = 169.3 + 3.2 - 7.6 - 18.4 = 146.6 KN/m

Shear strength $\varphi Vn = \varphi * 0.17 * (f'c)^{1/2} * 10 * d$ ACI Eq. (11-3)

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φ Vn = 0.75 * 0.17 * (35)½ * 10 * 63.8 = 469.6 KN/m > Vu = 97.6 KN/m OK

Use D25 @ 15.0 cm As = 32.73 cm²/m $\rho = As / b\,d$ = 32.73 / (100 * 63.8) = 0.0051

Bending strength $\varphi Mn = \varphi * d^2 * fc * q * (1 - 0.59 * q)$ ACI 10.2.7

φ Mn = 0.90 * 63.8² * 35.0 * 0.062 * (1 - 0.59 * 0.062) = 760.1 KN-m/m > Mu = 146.6 KN-m/m OK

Cover factor = $Min\ (2.5, (Cover + db/2, Spacing / 2) / db)$ = Min (2.5, (5.0 + 2.50 / 2, 15.0 / 2) / 2.50) = 2.5

Straight $Ld = Max\ (30, fy / 1.1 / (fc)^{½} * Size * Location / Cover * db * Ratio$ ACI Eq. (12-1)

= Max (30, 420.0 / 1.1 / (35)½ * 1.0 * 1.0 / 2.5 * 2.50 * 0.19) = 64.5 cm

Hooked $Ldh = 0.24 * fy / (fc)^{½} * db * 0.7$ = 0.24 * 420.0 / (35.0)½ * 2.50 * 0.7 * 0.19 = 29.8 cm ACI 12.5

Dev. length at heel side = $Ftg - Toe - Cover$ = (3.75 - 1.00) * 100 - 5.0 = 270.0 cm > 64.5 cm OK

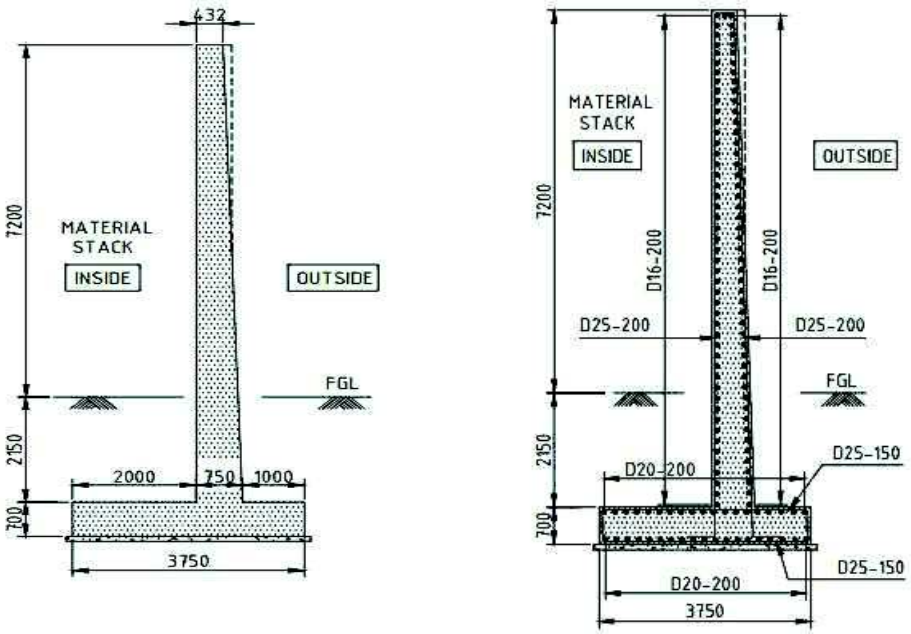
Dev. length at toe side = $Toe - Cover$ = 1.00 * 100 - 5.0 = 95.0 cm > 64.5 cm OK

$As\ min = Min\ (Max\ ((1.4 / fy, 0.25 * \sqrt{fc} / fy) * b * d), 1.33 * As\ req)$ ACI 10.5

= Min (Max ((1.4 / 420, 0.25 * (35.0)½ / 420) * 100 * 63.8, 1.33 * 5.5) = 8.2 cm²/m

$As\ S\&T = 0.0018 * b * t$ = 0.0018 * 100.0 * 70.0 = 12.60 cm²/m ACI 7.12.2.1

Min. steel area ratio = $As\ min / As$ = 12.60 / 32.73 = 0.38 < 1.0 OK



RETANING WALL TYPE - II

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REFERENCES

[1] Saudi concrete structures Code; SBC 304-2018

[2] Building Code Requirements for Structural Concrete; ACI 318-14

[3] Minimum Design Loads and Associated Criteria for Buildings and Other Structures; ASCE 7

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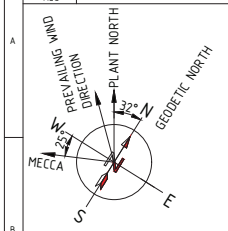
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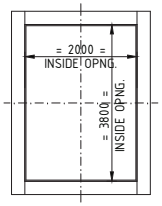
ATTACHMENT 1 MECHANICAL INPUT DRAWINGS

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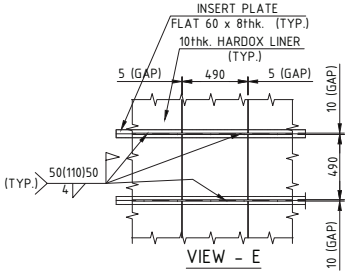
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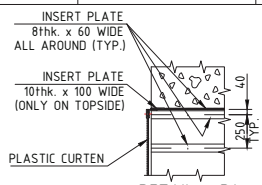




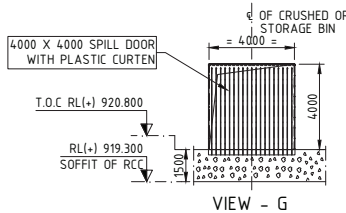
VIEW C-C
SCALE 1:40



VIEW - E
(TYP. LINER FIXING DETAILS ON VERTICAL RCC WALL)
SCALE 1:20



DETAIL - D4
SCALE 1:20

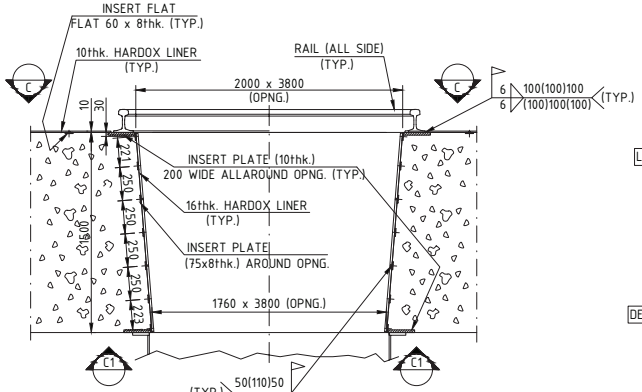


VIEW - G
SCALE 1:125

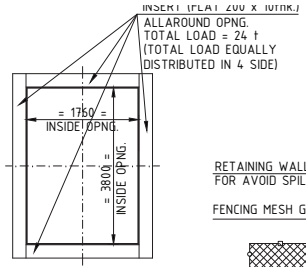
ABBREVIATION	
W.P.:	WORKING POINT
℄:	CENTER LINE
DN:	DOWNWARD
UP:	UPWARD
CRS.:	CENTERS
thk.:	THICKNESS
RCC:	REINFORCED CEMENT CONCRETE
deg.:	DEGREE
B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT

STORAGE MATERIAL CHARACTERISTICS		
PARTICULARS		SPECIFICATION
STORAGE MATERIAL		CRUSHED GOLD ORE
LUMP SIZE	(mm)	(-)10
BULK DENSITY	(t/m³)	1.6 (FOR VOLUME CALCULATION)
		1.9 (FOR STRUCTURAL CALCULATION)
MOISTURE CONTENT	(%)	2
ANGLE OF REPOSE	(deg.)	37
STORAGE PERIOD	(day)	1
STORAGE CAPACITY	(m³)	~9600 (INCLUDING DEAD VOLUME)
CHEMICAL PROPERTIES OF GOLD ORE :		REFER PROCESS DESIGN CRITERIA DOC. NO. MD-216-3000-PR-CRT-0001
DEAD VOLUME TO BE CLEARED PERIODICALLY TO AVOID CAKE FORMATION		

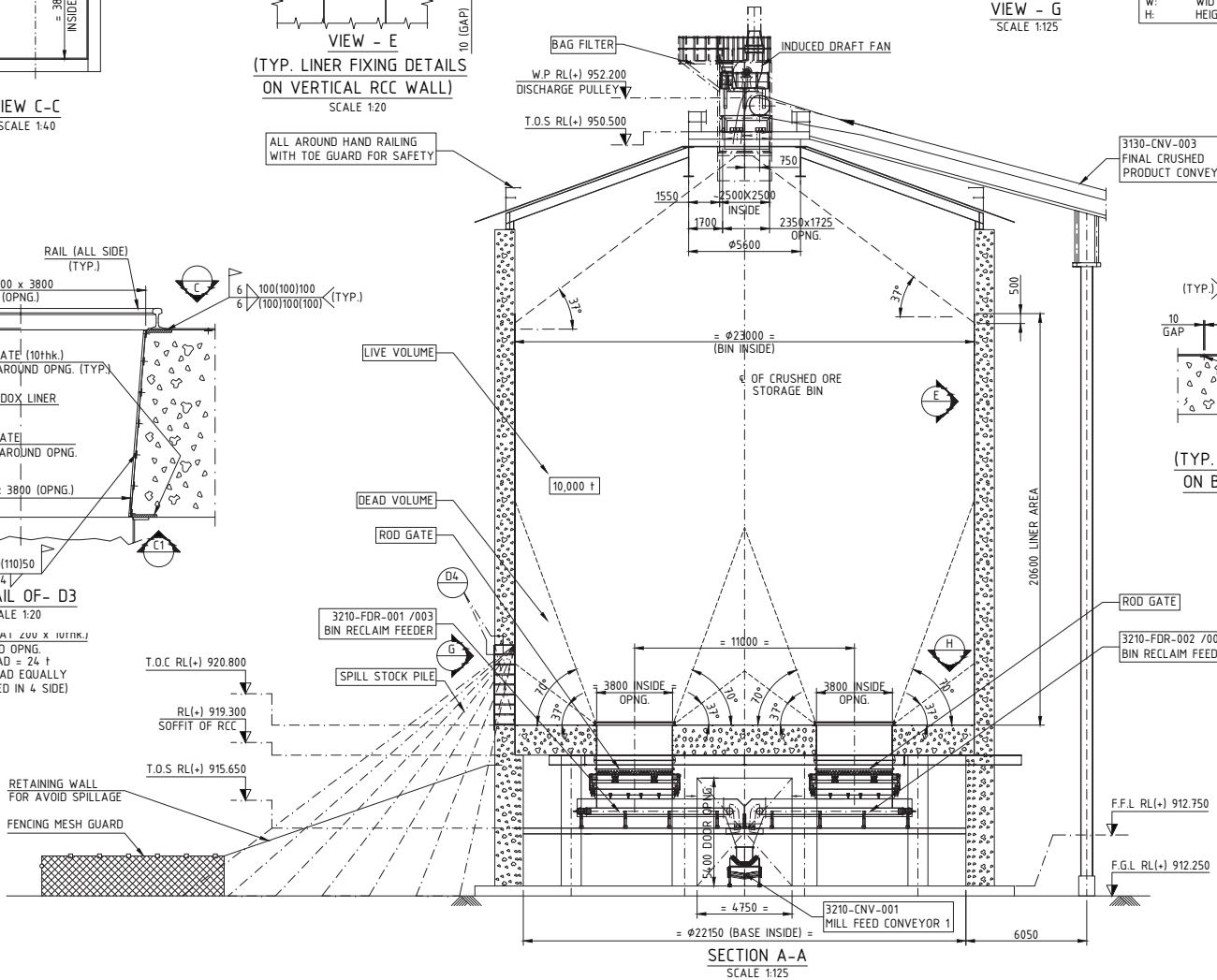
NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.



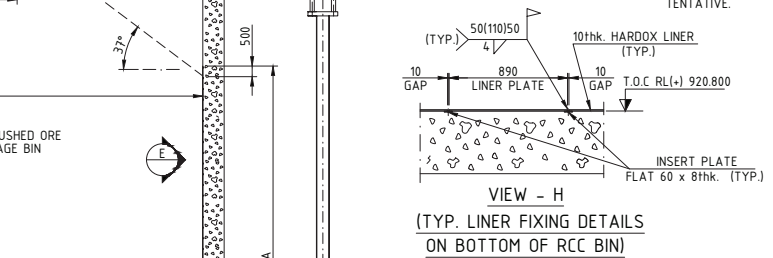
DETAIL OF- D3
SCALE 1:20



VIEW C1-C1
SCALE 1:40



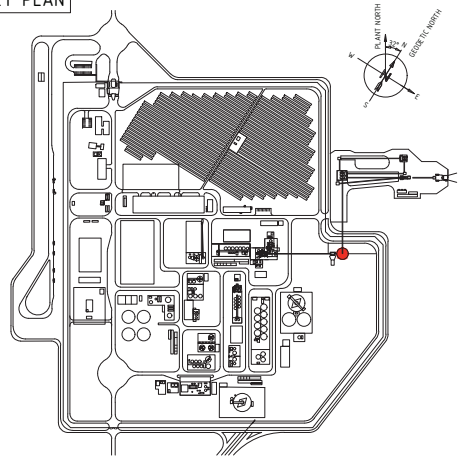
SECTION A-A
SCALE 1:125



VIEW - H
(TYP. LINER FIXING DETAILS ON BOTTOM OF RCC BIN)
SCALE 1:20

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN								
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION								
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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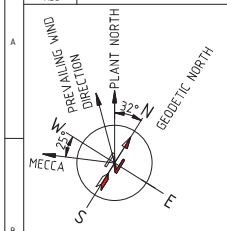
MA'ADEN معادن
Gold Amline Mining Company

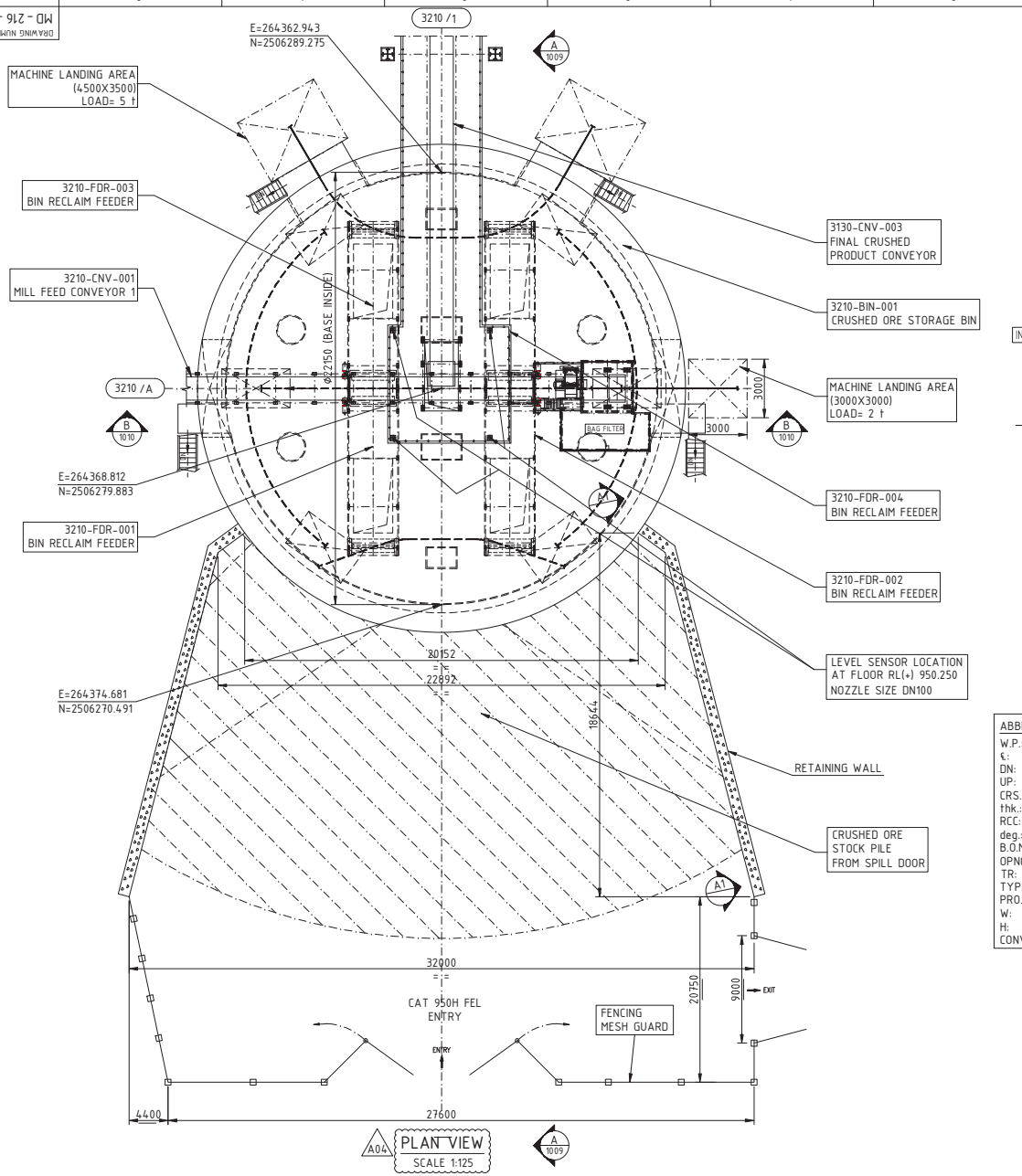
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LARSEN & TOUBRO DRAWING NUMBER	REVISION
O19045_100_H_G_16126_002	0
OUTOTEC DRAWING NUMBER	DOCUMENT ID / REVISION
900707 _GAU01 _US01_MTL01 _32114	OU100669133 / 0

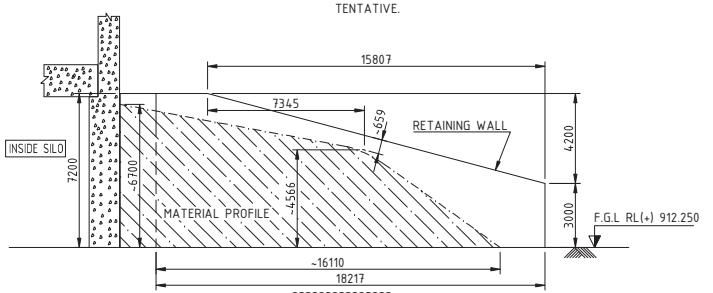
DRAWN	HKD	18/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	20/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	20/02/2020	GRINDING & CLASSIFICATION
APPROVED	BGR	21/02/2020	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION
EMPLOYER			
SCALE	1:125, 1:40, 1:20	A1	DRAWING NUMBER: MD-216-3210-EG-ME-ELV-1009 / REV: A01

DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1003 REV A04





NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.

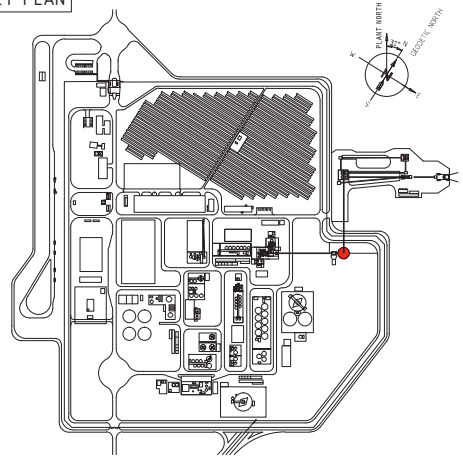


- CONCRETE FLOOR
- CHECKERED PLATE
- GRATING
- BRICK WALL
- CONCRETE WALL

ABBREVIATION

W.P.:	WORKING POINT
℄:	CENTER LINE
DN:	DOWNWARD
UP:	UPWARD
CRS.:	CENTERS
thk.:	THICKNESS
RCC:	REINFORCED CEMENT CONCRETE
deg.:	DEGREE
B.O.M.R:	BOTTOM OF MONORAIL
OPNG.:	OPENING
TR:	TRESTLE
TYP.:	TYPICAL
PROJ:	PROJECTED
W:	WIDTH
H:	HEIGHT
CONV.:	CONVEYOR

KEY PLAN



DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A04	22/09/2020	ISSUED FOR DETAIL DESIGN & REVISED AS MARKED	HKD	BRD	BD	BGR	
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A03	10/05/2020	ISSUED FOR DETAIL DESIGN	HKD	BRD	BD	BGR	
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN	A02	10/04/2020	ISSUED FOR APPROVAL & REVISED AS PER EMPLOYER	HKD	BRD	BD	BGR	
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE STORAGE BIN & TRANSFER - SECTION			COMMENTS DT. 20.03.2020					
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE STORAGE BIN & TRANSFER - ELEVATION	A01	21/02/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32103 — DOCUMENT ID: OU100669117 — REVISION: 3

DRAWN	HKD	18/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	20/02/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	20/02/2020	GRINDING & CLASSIFICATION
APPROVED	BGR	21/02/2020	CRUSHED ORE STORAGE BIN & TRANSFER - PLAN
EMPLOYER			
SCALE	1:125	A1	DRAWING NUMBER: MD-216-3210-EG-ME-LPX-1003 — REV: A04

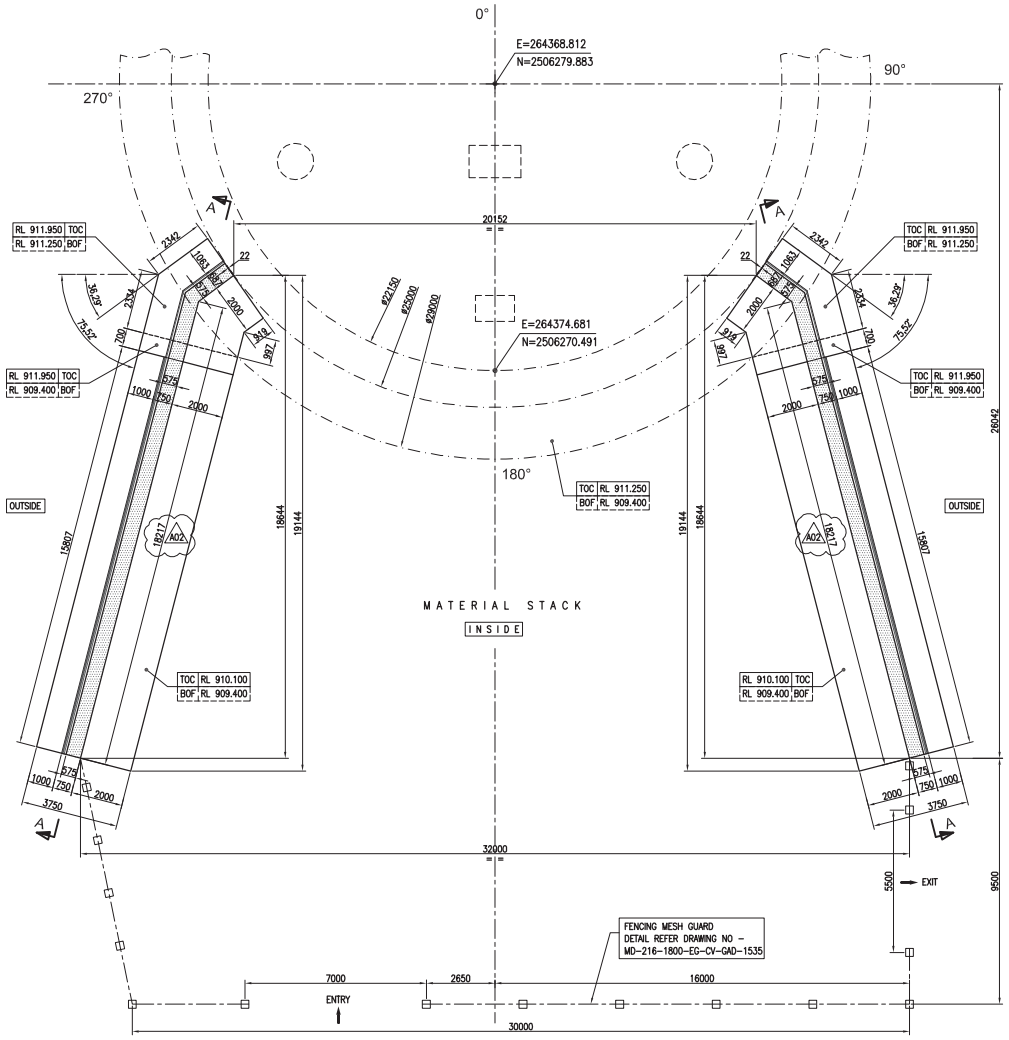
ATTACHMENT 2 CIVIL DRAWINGS

PROPRIETARY & CONFIDENTIAL

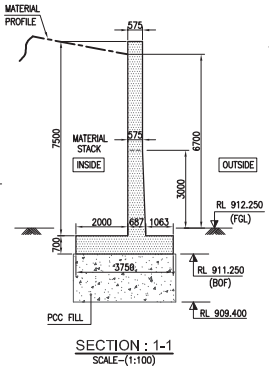
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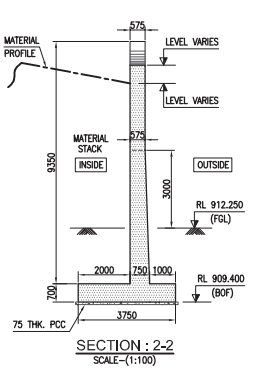
DRAWING NUMBER MD-216-3210-EG-CV-GAD-2001 REV A02



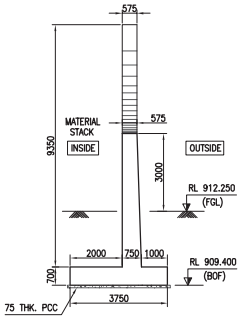
PLAN AT RL 909.400 (BOF)
SCALE-(1:100)



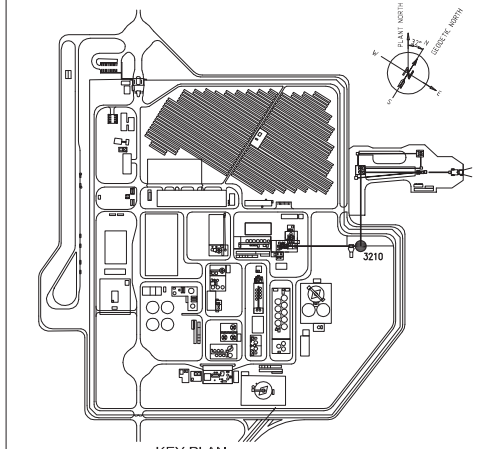
SECTION : 1-1
SCALE-(1:100)



SECTION : 2-2
SCALE-(1:100)



VIEW : 3-3
SCALE-(1:100)



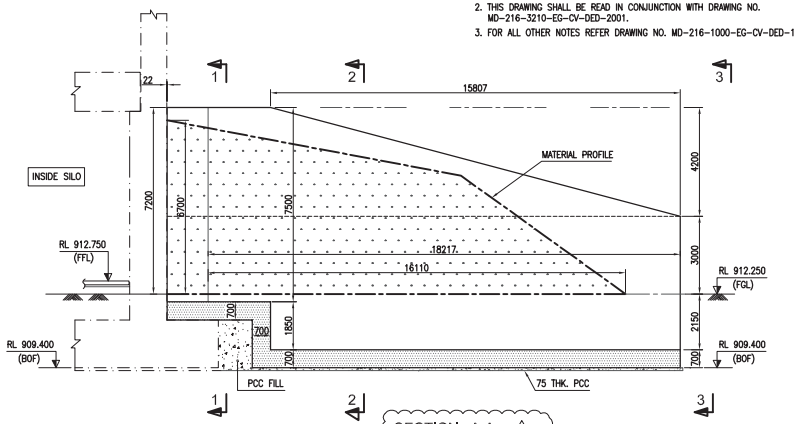
KEY PLAN
SCALE-(1:1000)

LEGEND :-

BOF = BOTTOM OF FOOTING
FGL = FINISHED GROUND LEVEL
FFL = FINISHED FLOOR LEVEL
TOC = TOP OF CONCRETE
PCC = PLAIN CEMENT CONCRETE
TYP = TYPICAL
MKD = MARKED
℄ = CENTRE LINE
THK = THICK
ø = DIAMETER
R = RADIUS
RL = REDUCED LEVEL

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES. UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING NO. MD-216-3210-EG-CV-DED-2001.
3. FOR ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001, 1003 & 1004.



SECTION : A-A
SCALE-(1:100)

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3210-EG-ME-ELV-1010	CRUSHED ORE BIN-SECTION								
MD-216-3210-EG-ME-ELV-1009	CRUSHED ORE BIN-ELEVATION								
MD-216-3210-EG-ME-LPX-1006	CRUSHED ORE BIN-PLAN								
MD-216-3210-EG-ME-LPX-1005	CRUSHED ORE BIN-PLAN	A02	08/10/2020	REVISED AS PER EMPLOYER COMMENTS	ABH	UR	SVK	AG	
MD-216-3210-EG-ME-LPX-1004	CRUSHED ORE BIN-PLAN			ON 25/09/2020 AS MARKED A02					
MD-216-3210-EG-ME-LPX-1003	CRUSHED ORE BIN-PLAN	A01	20/08/2020	ISSUED FOR REFERENCE	ABH	UR	SVK	AG	

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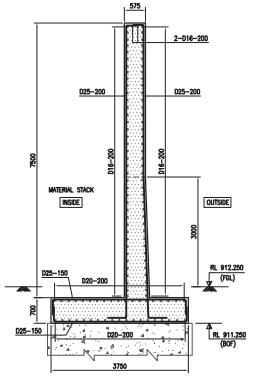
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LARSEN & TOUBRO DRAWING NUMBER: 019045_450_C_G_24377_001 — REVISION: 1
OUTOTEC DRAWING NUMBER: 900707__GAU01__US01__MTL01__32142 — DOCUMENT ID: OU100692098 — REVISION: 1

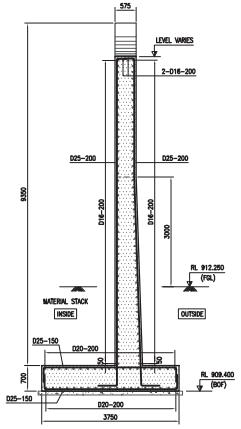
DRAWN	ABH	18/08/2020
CHECKED	UR	19/08/2020
ENGINEER	SVK	18/08/2020
APPROVED	AG	20/08/2020
EMPLOYER		
SCALE	1:100	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
GRINDING & CLASSIFICATION
CRUSHED ORE STORAGE BIN - RETANING WALL

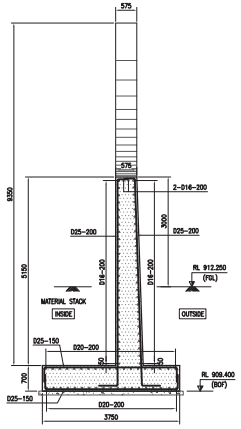
DRAWING NUMBER: MD-216-3210-EG-CV-GAD-2001 — REV: A02



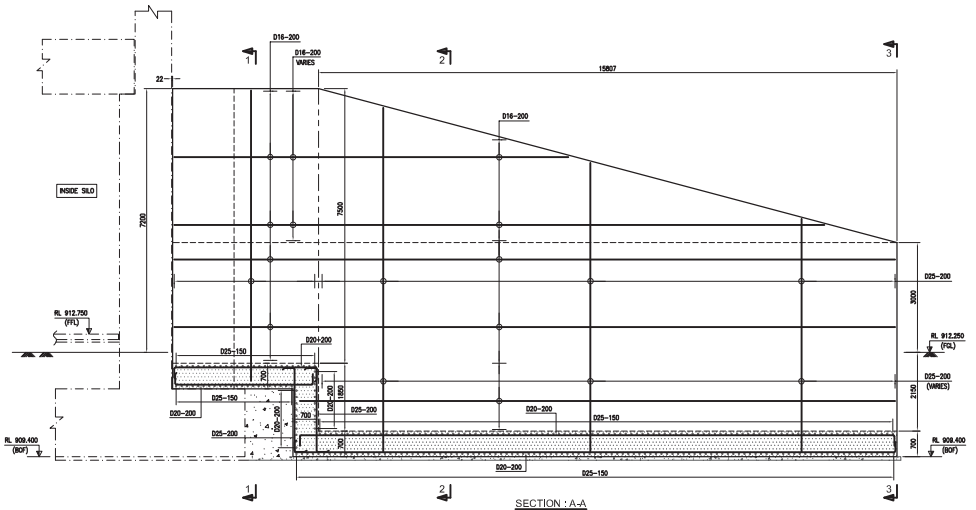
SECTION : 1-1
SCALE-(1:50)



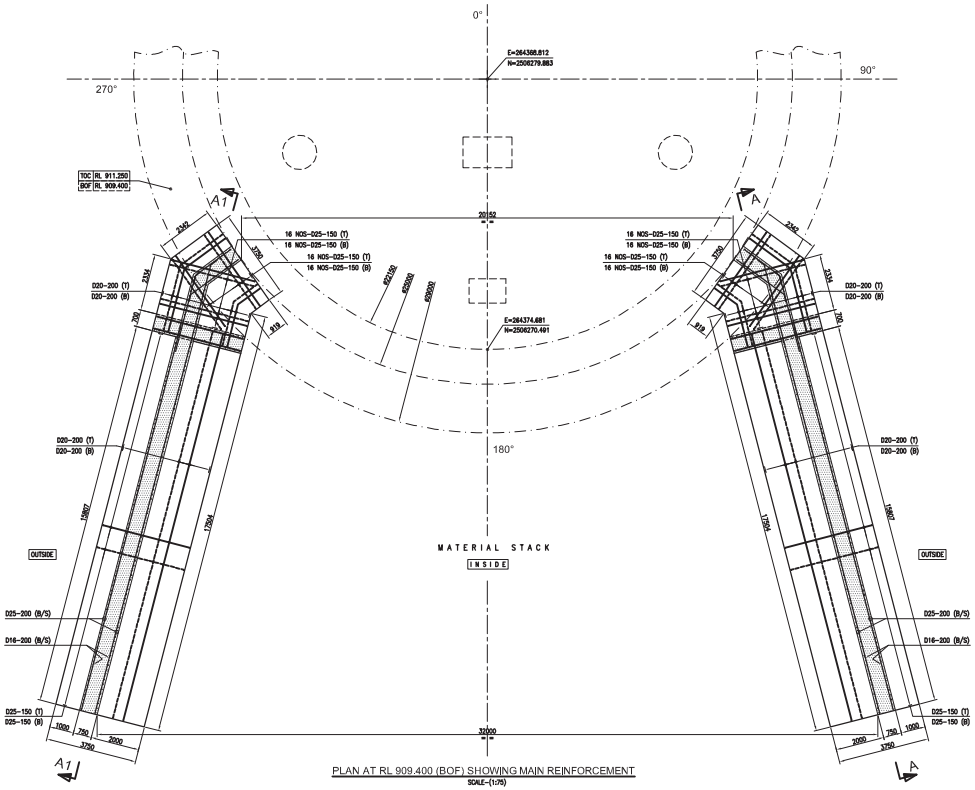
SECTION : 2-2
SCALE-(1:50)



SECTION : 3-3
SCALE-(1:50)



SECTION : A-A
SECTION : A1-A1 (OPPOSITE HAND)
SCALE-(1:50)



PLAN AT RL 909.400 (BOF) SHOWING MAIN REINFORCEMENT
SCALE-(1:75)

LEGEND:-

TOC = TOP OF CONCRETE
FFL = FINISHED FLOOR LEVEL
FGL = FINISHED GROUND LEVEL
BOF = BOTTOM OF FOOTING
PCC = PLAIN CEMENT CONCRETE
TYP = TYPICAL
MKD = MARKED
℄ = CENTRE LINE
ø = DIAMETER
R = RADIUS
B/S = BOTH SIDE
T = TOP
B = BOTTOM
RL = REDUCED LEVEL
= TOP REINFORCEMENT
= BOTTOM REINFORCEMENT

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS AND CO-ORDINATES ARE IN METRES, UNLESS OTHERWISE STATED.
2. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH DRAWING NUMBER : MD-216-3210-EG-CV-GAD-2001.
3. FOR ALL OTHER NOTES REFER DRAWING NO. MD-216-1000-EG-CV-DED-1001, 1003 & 1004.
4. GRADE OF CONCRETE SHALL BE C35.
5. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420 MPa)
6. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION CONSIDER 800KN/ SQ. M
7. IF ANY PART OF FOUNDATION COMES ON BACKFILL ZONE OF ADJACENT FOUNDATION OR ANY LOOSE SOIL ARE ENCOUNTERED BELOW FOUNDATION LEVEL THE SAME SHALL BE REMOVED AND FILLED WITH PCC.
8. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS :
   a) RAFT : 75 mm (BOTTOM & SIDE)
   b) RAFT : 50 mm (TOP)
   c) WALL : 50 mm
   a) COLUMN : 50 mm
9. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40xBAR DIAMETER.

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-3210-EG-PE-ELY-1010	CRUSHED ORE BIN-SECTION
MD-216-3210-EG-PE-ELY-1009	CRUSHED ORE BIN-ELEVATION
MD-216-3210-EG-PE-LPX-1006	CRUSHED ORE BIN-PLAN
MD-216-3210-EG-PE-LPX-1005	CRUSHED ORE BIN-PLAN
MD-216-3210-EG-PE-LPX-1004	CRUSHED ORE BIN-PLAN
MD-216-3210-EG-PE-LPX-1003	CRUSHED ORE BIN-PLAN

REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EPP
A02	08/09/2020	REVISED AS PER EMPLOYER COMMENTS ON 25/09/2020	ABH	UR	SVK	AG	
A01	28/08/2020	ISSUED FOR REFERENCE	ABH	UR	SVK	AG	

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LARSEN & TOUBRO DRAWING NUMBER		REVISION
019045_450_C_D_24378_001		1
OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
900707__GAU01__US01__CTB17__32108	OU100692099	1
DRAWN ABH 22/08/2020	MANSOURAH - MASSARAH GOLD PROJECT	
CHECKED UR 27/08/2020	DETAIL DRAWING	
ENGINEER SVK 22/08/2020	GRINDING & CLASSIFICATION	
APPROVED AG 28/08/2020	CRUSHED ORE STORAGE BIN - RETAINING WALL	
EMPLOYER		
SCALE 1:50, 1:25	A0	DRAWING NUMBER MD-216-3210-EG-CV-DED-2001 REV A02